

VLC- BASED PASSIVE POWERED SENSING AND CONTROL MODULE FOR INDOOR FARMING

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VLC PASSIVE POWERED SENSING AND CONTROL MODULE FOR INDOOR FARMING

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by

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Declaration

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We assure that any work we submit with our names on it will reflect our own ideas and effort. We will properly cite all material that is not our own. We understand that there is no acceptable excuse for committing plagiarism.

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Abstract

Indoor farming requires low-maintenance sensing and control nodes to monitor plant growth conditions and actuate environmental control devices while reducing wiring complexity and battery replacement. This research presents a VLC based passive powered control node for indoor farming, follows a Light-based Internet of Things (LIoT) approach. Visible Light Communication (VLC) is used as the downlink method to transmit control commands from a modulated light source to the node. A solar panel is used as the optical receiver as well as the energy harvesting element, following the Simultaneous Lightwave Information and Power Transfer (SLIPT) concept. The harvested energy is stored in a supercapacitor through an energy harvesting module and regulated to power the low-power control electronics. The AC component generated by the modulated light is extracted from the solar panel output, filtered, conditioned, and decoded by an MSP430 microcontroller. Binary Frequency Shift Keying (Binary FSK) was implemented as the VLC modulation method due to its suitability for robust low-data-rate command transmission. The node integrates four sensors to measure five indoor farming parameters: air temperature, relative humidity, soil moisture, soil temperature and light intensity. It also provides control signals for four external actuator channels, while the actuator power is supplied separately to maintain passive powered operation of the control node. An IR uplink using NEC-based encoding with CRC checking was implemented for acknowledgement and sensor/status data transmission. Experimental VLC testing achieved a data rate of 200 bps with a packet error rate of 0.90% in 1.2 m distance, demonstrating the feasibility of solar-panel-based VLC reception for low data rate indoor farming control. The developed system shows that a passive-powered LIoT control node can support sensing, optical command reception, actuator control, and uplink feedback.

Keywords: VLC, SLIPT, binary - FSK, Passive Powered Control Node, Indoor Farming.

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List of abbreviations

AC – Alternating Current

ADC – Analogue to Digital Converter

ASK – Amplitude Shift Keying

BER – Bit Error Rate

BLE – Bluetooth Low Energy

CO_2 – Carbon Dioxide

CRC - Cyclic Redundancy Check

DC – Direct Current

FSK – Frequency Shift Keying

GPIO – General Purpose Input/Output

I^2C – Inter- Integrated Circuit

IOT – Internet of Things

IR - Infrared

LED – Light Emitting Diode

LIOT – Light Based Internet of Things

LPM – Low Power Mode

MCU – Microcontroller Unit

MOSFET – Metal- Oxide- Semiconductor Field-Effect Transistor

MPP – Maximum Power Point

MPPT – Maximum Power Point Tracking

NEC – NEC Infrared Transmission Protocol

OOK – On Off Keying

PER – Packet Error Rate

PMU – Power Management Unit

PPFD - Photosynthetic Photon Flux Density

PWM- Pulse Width Modulation

RF – Radio Frequency

RX – Receiver

SLIPT – Simultaneous Lightwave and Power Transfer Techniques

TX - Transmitter

UART - Universal Asynchronous Receiver/Transmitter

VLC – Visible Light Communication

CHAPTER 1. Introduction

1.1 Background and Research Area

1.1.1 Indoor Farming and Controlled Environment Agriculture

Indoor farming is a controlled environment agriculture approach in which crops are cultivated inside enclosed or semi-enclosed spaces by regulating the growing conditions around the plants. Unlike open-field farming, indoor farming reduces direct dependence on external weather, seasonal changes and land availability. Indoor farm environments use artificial lighting, irrigation, ventilation, humidification, cooling and electronic monitoring to support stable crop production. Greenhouses are widely used in modern agriculture because they allow crops to be grown under adverse climatic conditions, and by regulating temperature, humidity, light, and carbon dioxide levels, farmers can improve productivity and crop quality [1].

The main purpose of indoor farming is to maintain suitable environmental conditions for plant growth. Important parameters include air temperature, relative humidity, growing-medium moisture and temperature, airflow, carbon dioxide concentration and light intensity. Temperature affects plant respiration, transpiration, enzyme activity and growth rate. Relative humidity influences transpiration and water movement between the plant and surrounding air. Growing-medium moisture and temperature affect water and nutrient availability in the root zone. Light intensity affects photosynthesis and plant development. Therefore, indoor farming systems require continuous observation of the growing environment so that suitable control actions can be taken when the measured conditions change.

In practical indoor farming systems, sensing and actuation are closely connected. Sensors are used to observe the growing environment, while actuators are used to correct or maintain the required conditions. Recent greenhouse sensing research states that maintaining ideal growing conditions for intensive production requires precise control of conditions such as irrigation, airflow, and light, also this requires arrays of sensors and actuators for automatic climate control [2].

Artificial lighting is an important part of many indoor farming systems because it supplements or replaces sunlight. LED-based lighting is mostly used in greenhouses and indoor farms because LEDs are efficient, controllable, and suitable for integration into automated growing systems.

Environmental conditions inside an indoor farming area are not always uniform. The temperature, humidity, moisture level and light exposure can vary between plant trays, heights and positions inside the growing area. Airflow distribution, plant canopy structure, distance from the light source and irrigation layout can create local differences in the plant environment. Therefore, relying on a single centralized sensor may not provide enough information for the whole farming area. Distributed sensing and control nodes are useful because they can collect local environmental data and support localized control actions.

The deployment of distributed nodes also requires attention to power consumption, communication method and maintenance. Environmental sensing and monitoring systems often require dense deployment of sensors to achieve accurate data acquisition and energy-autonomous devices, efficient low power communication protocols and sustainable design approaches are important for reliable operation. In indoor farming, these requirements are important because nodes may be placed close to plants, irrigation lines, humid areas, or locations that are not convenient for frequent maintenance.

For this type of application, a control node should be able to read environmental parameters and provide control outputs to actuators. The measured parameters may include air temperature, relative humidity, growing medium moisture, growing medium temperature and light level as a practical indication of illumination condition. High power actuators such as fans, pumps, solenoid valves, humidifiers, and cooling modules usually require a separate power.

1.2 Comprehensive Literature Survey

1.2.1 Visible light communication (VLC)

1.2.1.1 VLC-Based Communication in Greenhouse and Indoor Farming Systems

Recent developments in controlled environment agriculture have created interest in using optical communication for greenhouse and indoor farming automation. Since these environments already use artificial lighting for plant growth, visible light can be used not only for illumination but also for data transmission. Morales et al. [1] presented a solar-panel-based VLC system for greenhouse applications and highlighted that greenhouse environments require continuous monitoring of parameters such as temperature, humidity, light, and carbon dioxide to improve productivity and crop quality. Their work experimentally evaluated a VLC system using a small solar panel as the optical receiver, focusing on bit error ratio, received optical power, humidity variation, and transmitter–receiver distance. The study showed that solar-panel-based VLC is practical for short-range and low-data-rate greenhouse communication, although link performance decreases as the distance increases and environmental conditions become more challenging [1].

Guo et al. [4]proposed a greenhouse sensing system using a chirp-based VLC downlink, RF backscatter uplink, and a solar panel for both energy harvesting and signal reception. Their work is highly relevant to this research because it directly connects greenhouse sensing, VLC downlink communication, solar energy harvesting, and low-power node operation. The authors explain that maintaining ideal growing conditions requires control of irrigation, airflow, and light using arrays of sensors and actuators. They also show that LED grow lights can provide both plant illumination and a communication downlink for solar-powered IoT nodes [4]. This supports the suitability of optical communication links in controlled farming environments where lighting infrastructure is already available.

VLC is attractive in indoor farming because it can reduce dependence on radio-frequency communication for local command transmission. In VLC, information is transmitted by modulating the intensity of an LED light source. Since LEDs are already used in indoor farms and greenhouses, VLC can be implemented without introducing a completely separate communication medium. However, the modulation method must be selected carefully because greenhouse and indoor farming environments can contain optical noise from sunlight, grow lights, reflections, and other artificial light sources. Guo et al. [4] noted that OOK is simple and low-cost but highly susceptible to ambient noise, and their chirp-

based approach was introduced to improve noise immunity and reduce error rates in greenhouse VLC.

1.2.1.2 Modulation Techniques for Robust VLC Communication

The selection of modulation technique affects VLC reliability, circuit complexity, and receiver power consumption. OOK is one of the simplest VLC modulation methods because the LED is switched between ON and OFF states. However, OOK can be sensitive to ambient light noise and amplitude variations. Guo et al. [4] identified this limitation in greenhouse VLC and proposed chirp-based VLC to improve noise immunity in greenhouse environments, where low cost and robustness are more important than high data rate.

Binary Frequency Shift Keying (Binary FSK) is another modulation method that can improve robustness by representing digital bits using two different frequencies instead of relying only on amplitude level. Beguni et al. [2] experimentally demonstrated a VLC system based on Binary FSK modulation and compared its noise resilience with a standard OOK VLC system. Their results showed improved optical noise resilience under both direct and indirect incandescent light exposure compared with OOK [2].

This supports the use of Binary FSK in the present research, where the VLC downlink is intended for low-data-rate but reliable command transmission. Since the control node receives short command frames rather than large data streams, robustness is more important than high throughput. Therefore, Binary FSK is a suitable modulation approach for the proposed passive-powered indoor farming control node.

1.2.1.3 Light-Based IoT (LIOT)

Light-based Internet of Things (LIoT) is a research area where visible and non-visible optical links are used for communication between low-power embedded nodes and gateways. Although the term IoT is often associated with internet-connected systems, in this research context LIoT refers to embedded sensing or control nodes that communicate using optical links. Landivar et al. [5] investigated batteryless BLE and LIoT sensor nodes for reliable environmental sensing and described LIoT nodes as systems that can use different parts of the optical spectrum for uplink and downlink communication, such as visible light and infrared.

Landivar et al. [5] showed that VLC and IR can be combined to create bidirectional optical communication. In their LIoT node data exchange, the node uploads its ID and illumination value through IR, the gateway sends requests through VLC, and the node sends the requested sensor data back to the gateway. This supports the communication structure used in the present research, where VLC is used as the downlink for command reception and IR is used as the uplink for acknowledgement and sensor/status data transmission.

Landivar et al. [5] also discussed the limitations of RF communication in dense IoT deployments. Although BLE is suitable for low-power short-range communication, BLE and other RF technologies may experience congestion and interference when many devices are deployed. VLC provides an alternative communication method by using the light spectrum, while LIoT can improve reliability and node efficiency by combining visible and infrared communication links without mutual RF interference [5]. This is relevant to indoor farming because several sensor and actuator nodes may be distributed inside a limited growing area.

Liu et al. [6] proposed Retro-VLC, a low-power duplex visible light communication system for battery-free mobile and sensor devices. In their system, the downlink is provided by an illuminating LED, while the uplink is achieved by reflecting and modulating the incoming light using a retro-reflector and an LCD shutter. The tag side also uses a solar cell to harvest energy from the illumination source, enabling battery-free operation. Their prototype demonstrated bidirectional optical communication with 10 kbps downlink and 0.5 kbps uplink over a distance of up to 2.4 m [6]. Although the uplink method used in Retro-VLC is different from the IR uplink used in the present research, the work demonstrates the feasibility of low-power bidirectional optical communication for batteryless sensing nodes.

1.2.2 IR Feedback Uplink

Optical wireless communication can support short-range data transfer without using radio-frequency communication. This is suitable for the proposed control node because the indoor farming system already includes a controlled optical source for illumination and VLC downlink transmission. Therefore, visible light can be used for downlink command transmission, while infrared communication can be used as an uplink feedback path from the control node.

A bidirectional optical communication system requires both downlink and uplink paths. Pérez-Nicoli et al. [7] reviewed uplink wireless transmission methods for bidirectional VLC systems and explained that the uplink requires a suitable return channel from the user device or receiver node. This supports the communication structure used in the present research, where the grow light unit sends downlink commands using VLC and the control node sends feedback using infrared communication.

Infrared communication is suitable for the uplink because it does not visibly affect the grow-light output. Alsulami et al. [8] proposed an infrared uplink design for VLC systems and showed that IR transmission can be used as the return path in a VLC-based system. Chen et al. [9] also studied wireless infrared-based LiFi uplink transmission and analysed practical issues such as link blockage and random device orientation. These studies show that IR uplink communication is a practical approach for optical wireless systems, although the link direction and receiver placement must be considered during system design.

The proposed control node does not require a high data-rate IR uplink. The uplink is mainly required for acknowledgement, sensor information and actuator status feedback. Therefore, a custom IR packet structure was designed using NEC-style timing. NEC-style infrared communication uses a 38 kHz carrier with pulse distance timing, which is compatible with common demodulating IR receiver modules [10]. In this research, the NEC-style timing structure is adapted for a custom feedback frame rather than a standard remote-control command. This allows the control node to transmit application specific feedback while keeping the IR receiver circuit simple and low cost.

The use of an IR feedback link improves the practical operation of the VLC-based control node because the external system can confirm whether a command was received. This is important for actuator control applications because the system should avoid acting on unconfirmed or corrupted optical commands. Therefore, the IR uplink complements the VLC downlink and supports RF-free bidirectional optical communication at the control-node level.

1.2.3 Simultaneous Lightwave Information and Power Transfer Techniques

Simultaneous Lightwave Information and Power Transfer (SLIPT) is the optical equivalent of using the same medium for both communication and energy transfer. Pan et al. [11] reviewed SLIPT techniques and explained that visible light communication can provide data transmission while the energy of the optical signal can be harvested by energy-constrained terminals such as indoor sensors and portable devices. They also discussed SLIPT receiver design methods based on time, signal components, and photoelectric converters [11].

SLIPT is especially suitable for low power nodes because light can act as both the communication carrier and the energy source. Pan et al. [11] explained that LED lamps can be used as energy sources and that large area photodetectors can be used as receivers to collect optical energy. In an indoor environment, the received optical power depends strongly on the transmitter - receiver distance and receiver position because optical channel gain decreases with distance [11]. This is important for indoor farming because the control node position, solar panel orientation and distance from the light source can affect both harvested energy and VLC signal quality.

De Oliveira Filho et al. [12] developed a SLIPT-based IoT prototype using a large area solar panel to harvest energy and decode optical data signals. Their work explains that solar cells can be used to harvest energy and receive information signals. They also describe different SLIPT configurations, including time switching, power splitting, and space switching and compare them in terms of hardware complexity, real-time energy harvesting and information decoding, MPPT possibility and cost [12].

For the present research, the solar panel is used as a dual-purpose optical interface. The DC component supports energy harvesting, while the modulated AC component is extracted for VLC command reception. This approach is closely related to the signal component separation concept in SLIPT, where energy harvesting and information decoding are both obtained from the received light. The use of a solar panel as both an energy harvester and optical receiver reduces the need for an additional photodiode receiver and supports compact passive-powered node design.

Gutiérrez et al. [13] developed a batteryless VLC receiver node in which a solar panel functions as both the energy harvester and optical antenna. Their system uses a lightweight analog front-end to filter and digitize the solar panel signal for a low power processor. The processor manages operation according to stored energy level and VLC reception is synchronized with the harvester's open-circuit phase to reduce interference and improve signal quality [13]. Their prototype decoded 32-bit VLC frames at an optical clock of 800 Hz, consumed less than 2.8 mJ, and maintained sleep-mode power below 30 μW [13].

The work of Gutiérrez et al. [13] also shows a practical receiver architecture with a solar panel, energy harvester, storage capacitor, analog front-end, voltage divider, low-power processor, and main processor. Their analog front-end includes a high-pass filter and comparator, where the high-pass filter removes the DC component from the solar panel voltage and conditions the signal for MCU detection [13]. This supports the use of filtering and comparator-based detection in a low-power solar-panel VLC receiver.

1.2.4 Passive-Powered and Batteryless Sensor Nodes

Passive-powered and batteryless sensor nodes are designed to operate using harvested energy instead of relying on continuous wired power or frequent battery replacement. Landivar et al. [5] stated that sustainable IoT design must consider energy efficiency, energy autonomy, and reliable communication so that energy is not wasted on undelivered data. Their BLE and LIoT nodes use indoor light energy harvesting, low-power sensors, efficient system-on-chip devices, and active/sleep duty cycle adaptation based on available energy [5].

Energy storage is a major part of batteryless operation. Landivar et al. [5] described an energy harvesting unit with MPPT and a supercapacitor as an energy buffer. They also justified the use of supercapacitors due to high charge - discharge efficiency, high power density and long cycle life. In their experimental results, the BLE and LIoT nodes achieved bidirectional communication intervals of 19.3 s and 624.6 s respectively under energy-autonomous operation [5]. This shows that optical and low power communication systems can operate from harvested indoor light when the active and sleep cycles are designed carefully.

Gutiérrez et al. [13] further demonstrated that a batteryless VLC node can remain in a low power sleep state while waiting for sufficient stored energy, then execute VLC or BLE-related tasks without dropping below the harvester threshold. Their system consumed less than 0.030 mW in sleep mode and could reliably execute communication states from stored capacitor energy [13]. This supports the importance of low power firmware, energy aware operation and duty cycled control in passive-powered nodes.

1.2.5 Smart Agriculture Monitoring, Local Processing, and Control Reliability

Smart agriculture systems use embedded sensing, communication, processing and control technologies to improve the monitoring and management of crop-growing environments. In controlled environment agriculture, this is especially important because environmental parameters directly influence plant growth, water use, transpiration, photosynthesis, nutrient uptake, and crop quality. Since indoor farms, greenhouses, and polytunnels depend on artificial or semi-controlled environments, the ability to measure local environmental conditions and apply corrective actions is essential for stable plant production.

Jaliyagoda et al. [14]developed and evaluated a low-cost smart agriculture system for polytunnels using distributed sensor nodes, a controller node, a web server, and a dashboard. Their sensor nodes measured temperature, humidity, and light intensity, while the controller node operated greenhouse actuators such as a blower motor and misting system. This work shows that low-cost embedded hardware can support protected-agriculture monitoring and actuator control. It also highlights the importance of sensor placement, because measurements taken at different positions and heights inside a polytunnel can represent environmental variation more accurately than a single centralized sensor.

Smart greenhouse systems also require multi-sensor monitoring and reliable data processing to support automatic environmental control. Bicamumakuba et al. [15] reviewed smart greenhouse environment management and identified multi-sensor monitoring, intelligent control, and data processing as key elements of modern greenhouse systems. Their review shows that parameters such as temperature, humidity, soil moisture, light intensity, carbon dioxide and energy-related variables are commonly monitored for greenhouse control. They also highlighted practical challenges such as sensor noise,

calibration variation, missing data, interoperability issues and environmental disturbances, which can affect the reliability of control decisions.

Automatic actuation is a necessary extension of environmental monitoring. Sensor measurements are useful only when they can support suitable control actions such as ventilation, irrigation, humidification, cooling or valve operation. In controlled farming systems, these actuator functions help regulate the microclimate around plants and reduce unwanted variations in temperature, humidity, air circulation, and growing-medium moisture. Therefore, a practical control node should not only collect environmental data but also support actuator control decisions based on measured conditions.

Local processing is important in such systems because sending every sensor reading to a remote server or cloud platform can increase communication load, response time, and energy consumption. Alharbi and Aldossary [16] proposed an energy efficient edge-fog-cloud architecture for smart agriculture and explained that real time agricultural data such as temperature, soil moisture, soil acidity and irrigation information can be processed closer to the sensor layer. Their work shows that distributing processing across edge, fog, and cloud layers can reduce energy consumption, carbon emissions and network traffic compared with a traditional cloud-based approach [16]. Fernando et al. [17] also discussed edge-fog-cloud collaboration and showed that moving computation closer to the data source can improve performance and reduce energy use depending on the available processing resources and task requirements.

Reliable sensor data are also required for stable actuator control. Raw sensor readings can contain short term noise, sudden spikes, missing samples or values outside the expected physical range. If such readings are used directly, the controller may switch actuators unnecessarily or incorrectly. For example, a temporary false humidity reading could activate a humidifier, while a noisy soil moisture value could cause incorrect irrigation control. Therefore, filtering and validation methods are important in automatic environmental control systems. Bicamumakuba et al. [15] discussed filtering methods such as moving average filters, low pass filters, Kalman filters, adaptive filtering and other data processing methods for greenhouse sensor data. These methods help smooth sensor

readings, reduce noise effects, detect abnormal values and improve the reliability of control decisions.

For low power embedded control nodes, the filtering and processing method must be selected according to memory capacity, processing capability and energy budget. Advanced filtering methods may improve accuracy, but they can also increase computational complexity. Therefore, lightweight methods such as averaging, moving average filtering or exponential smoothing are more suitable for microcontroller-based nodes that operate from limited energy sources.

In the present research, these concepts are applied to a VLC-based passive-powered control node architecture. The proposed node measures air temperature, relative humidity, growing-medium moisture, growing-medium temperature, and light intensity. The MSP430 microcontroller performs local processing by reading sensor values, applying basic filtering, validating received VLC command frames and generating logic level actuator control signals. This approach reduces the need for continuous external processing, supports faster local response to environmental changes, and reduces unnecessary communication. Since the proposed node is passive powered, local processing and lightweight filtering are important for maintaining low power operation while improving actuator stability and control reliability.

1.2.6 Threshold Selection and Adaptability to High-Value Crops

Controlled environment agriculture is especially important for high-value crops because yield, quality, and resource efficiency directly affect economic performance. Strawberry is one example of a high-value horticultural crop that is widely cultivated under protected and controlled environments. Global strawberry production is about 10 million tons annually, with an estimated value of about US$22 billion from approximately 400,000 ha of cultivation area [3]. This shows that high-value crops can benefit from precise environmental monitoring and automated control systems that reduce manual intervention, improve resource use, and support stable production. The proposed VLC based passive-powered sensing and control node architecture can be adapted to different crops by changing the control thresholds.

Strawberry (*Fragaria* × *ananassa* Duch.) is a high value horticultural crop that is widely grown in protected and controlled environments. Its growth, flowering, fruit set, fruit size, and fruit quality are strongly affected by temperature, humidity, light, irrigation, and growing-medium conditions [18].

Temperature is one of the main parameters affecting strawberry growth and production. Greenhouse strawberry production guidelines report that a day temperature of about 75 °F, or 23.9 °C, is suitable for vegetative growth, while a night temperature of about 55 °F, or 12.8 °C, is used during fruiting. An average daily temperature of about 65 °F, or 18.3 °C, is suitable for flowering [18]. In another controlled-environment strawberry study, plants were maintained at 20 ± 2 °C during the day and 16 ± 2 °C at night [19]. These findings show that air-temperature monitoring is required in a strawberry growing system.

Relative humidity is also important for strawberry production. Lieten studied the effect of humidity on greenhouse-grown strawberry and reported that 65–75% relative humidity during the day was suitable for good yield, fruit size, and fruit set [20]. The same study reported that very high humidity could reduce fruit set, increase malformed fruit, reduce firmness, and increase fruit rot during storage [20]. Therefore, humidity control is required to avoid both dry and excessively humid conditions.

Light condition is another important requirement for strawberry cultivation. Greenhouse strawberry production guidelines report that sufficient daily light is required for greenhouse production, with a daily light integral of 20 - 25 mol m^{-2} day^{-1} recommended for strawberry cultivation. Day-neutral strawberry varieties can also perform well when 16 h of light is provided [18]. Controlled environment lighting research further shows that optimized LED lighting can improve strawberry growth and fruit production. Zou et al. [19] used red LEDs and full-spectrum white LEDs and reported that a 16 h optimized lighting treatment improved plant height, leaf width, and fruit weight compared with lower light treatments. This supports the need to monitor the lighting condition around the plant canopy.

Growing-medium moisture is also important because strawberry has a shallow root system and is sensitive to water deficit. Attar et al. [21] reported that water deficit can affect photosynthetic performance and reduce yield in strawberry cultivars. Therefore,

monitoring growing-medium moisture is necessary to support irrigation control and to reduce both water stress and excessive irrigation.

The root-zone condition must also be considered because the growing medium can have a different temperature from the surrounding air, especially when cooling or heating is applied near the plant tray. Therefore, air-temperature measurement alone is not sufficient for the proposed control node.

Multi-parameter monitoring is also supported by previous greenhouse strawberry monitoring research. Xie et al. [22] stated that strawberry growth is sensitive to environmental parameters such as temperature, humidity, light intensity, and gas concentration. Their monitoring system used multiple sensors to support greenhouse strawberry management [22]. This supports the multi-sensor approach used in the proposed control node, where air temperature, relative humidity, growing-medium moisture, growing-medium temperature, and light intensity are measured to support environmental control decisions.

Based on the strawberry literature, the control thresholds used in this research were selected to support air-temperature regulation, humidity regulation, irrigation control, growing-medium temperature monitoring, and light-condition monitoring. However, the proposed hardware architecture is not limited to strawberry. Since the threshold values are implemented in firmware, the same control-node platform can be adapted to other high-value controlled-environment crops by changing the threshold profile.

Saffron (*Crocus sativus* L.) is one example of another high-value crop that can benefit from this type of controlled environment monitoring and control system. Previous saffron studies show that growth, flowering, corm development, humidity, root-zone temperature, and artificial lighting conditions can influence saffron production [23] [24] [25]. Therefore, although strawberry was selected for the experimental threshold setting and actuator validation in this research, the proposed control node architecture can also be adapted in future work for saffron and other high-value crops grown under controlled conditions.

1.3 Research Problem

Indoor farming systems require continuous monitoring and actuation to maintain suitable environmental conditions for plant growth. In a practical setup, sensing points and actuator control points may be distributed across different plant rack trays, different heights, and different crop zones. Conventional wired sensing and control systems can support reliable operation, but the installation becomes complex when the number of nodes increases. Long wiring for sensors, actuator control lines, and power distribution increases cost, reduces flexibility, and makes maintenance difficult, especially in humid indoor farming environments. Wireless nodes can reduce cabling, but commonly used RF-based technologies such as Wi-Fi, BLE, Zigbee, and LoRa require additional radio hardware and consume power during communication.

Recent research has shown that RF-based low power communication methods such as BLE are useful for monitoring applications, but dense IoT deployments may still experience congestion and interference. VLC and LIoT have been proposed as alternatives or complementary technologies because they use the light spectrum for short-range wireless communication and can reduce dependence on RF channels [12]. In an indoor farming environment, the same artificial environment can be used for communication and energy harvesting.

Energy supply is another major challenge. A large number of distributed nodes cannot depend only on frequent battery replacement, and continuous wired power reduces the flexibility of the system. Batteryless and passive-powered nodes have been investigated for environmental sensing, where energy is harvested from indoor light and stored in an energy buffer such as a supercapacitor [12]. However, a control node for indoor farming has a more demanding role than a simple sensing node because it must support command reception, local sensing, decision-related processing, actuator control signal generation, and uplink feedback. Therefore, the node architecture must be designed carefully so that low-power electronics can operate from harvested energy while higher-power actuators are powered separately through an external driver circuit.

The use of a solar panel as both an energy harvester and a VLC receiver provides a possible solution to this issue. SLIPT research shows that solar cells can harvest energy from light

sources while also receiving information signals [4]. Solar panel-based VLC research also demonstrates that a solar panel can act as a dual-purpose optical interface, reducing additional receiver hardware and supporting low-power operation [13]. However, using a solar panel as a VLC receiver creates practical difficulties. The received signal contains both a DC component for energy harvesting and a small AC component for data reception. The solar panel also has limited frequency response due to its internal capacitance and low-pass behavior. Therefore, a suitable signal conditioning circuit is required to extract the communication signal without disturbing the energy harvesting function.

Communication reliability is also important in this research problem. A sensing node can sometimes tolerate a delayed reading, but an actuator control node must avoid corrupted or wrongly decoded commands. If an incorrect command is accepted, an actuator such as a fan, pump, solenoid valve, humidifier, or cooling device may operate incorrectly. Previous VLC studies commonly evaluate communication quality using BER, while a control-node application requires command-level reliability, such as packet error rate, frame validation, and acknowledgement. Therefore, the receiver must not only detect the VLC signal but also validate the received frame before performing a control action.

Another important issue is bidirectional communication. A VLC downlink allows the node to receive commands from the light source, but the system also needs a return path to confirm whether the command was received correctly and to transmit sensor/status data. LIoT research has shown that visible light can be used for downlink communication and infrared can be used for uplink communication in batteryless sensor nodes [12]. This is suitable for indoor farming because IR communication provides a simple optical feedback path without adding RF transmission hardware. However, for actuator control, the uplink should be used together with frame validation and acknowledgement so that the control system can confirm the command execution status.

Based on these considerations, the main problem addressed in this research is the lack of a low power, passive powered control node architecture for indoor farming that can reduce wiring and battery dependency while avoiding the need for RF communication at the node level. The control node must be able to harvest energy from light, receive control commands through a solar panel-based VLC receiver, validate received commands before

actuator operation, read local environmental sensors, generate control outputs for external actuator drivers, and transmit acknowledgement or sensor/status information through an infrared uplink.

This research problem leads to the need for a compact and low-power control node that integrates optical energy harvesting, solar-panel-based VLC reception, Binary FSK command decoding, local sensor interfacing, actuator control signal generation, and IR uplink feedback. The developed node should be suitable for indoor farming applications where low maintenance, reduced wiring, and reliable local control are required.

The reviewed literature shows that smart greenhouse monitoring, sensor filtering, actuator control, SLIPT, and solar-panel-based VLC have been studied in different contexts. However, these studies are generally presented as separate sensing, communication, energy-harvesting, or control solutions. They do not fully address the development of a compact passive-powered optical control-node architecture that combines optical energy harvesting, VLC downlink communication, local sensor processing, actuator control, and infrared uplink feedback for indoor farming.

Therefore, this research addresses this gap by developing a VLC-based passive-powered sensing and control node for strawberry-based controlled-environment cultivation. The node receives both energy and downlink data from the grow-light unit through a solar panel, measures local environmental parameters using multiple sensors, supports actuator-control decisions through local processing, and sends acknowledgement, sensor readings, and actuator status back to the grow-light unit through infrared communication.

Although strawberry was used as the practical crop case for threshold selection and actuator validation, the proposed architecture is not limited to strawberry. Since the control thresholds are implemented in firmware, the same control-node platform can be adapted to other high-value controlled-environment crops, including saffron, by modifying the threshold profile according to crop-specific requirements.

1.4 Justification and Originality of the Proposed Research

Most existing works focus mainly on communication demonstrations, sensing nodes, or energy harvesting techniques. In contrast, indoor farming requires a node that can perform

both sensing and actuator control. Therefore, this research focuses on developing a practical passive-powered control node that can receive optical commands, measure local environmental parameters, generate actuator control signals, and send feedback using RF-free communication at the node level.

The originality of this research lies in integrating VLC, SLIPT, IR communication and passive powering concepts into a single embedded control-node architecture for indoor farming. The developed system combines SLIPT, Binary FSK VLC downlink communication, NEC-based infrared uplink communication, CRC-based frame validation, local sensor processing, and external actuator control.

VLC is used as the downlink to transmit commands to the control node, while IR communication is used as the uplink for acknowledgement and sensor/status data transmission. This creates a bidirectional optical communication structure suitable for local indoor farming control. The use of CRC-based validation helps the system detect corrupted frames before accepting received commands or feedback data.

The solar panel is used as a dual-purpose device. It harvests energy from the available light and also acts as the VLC receiver. This follows the SLIPT concept, where light is used for both information transfer and energy harvesting [4]. The harvested energy powers the low-power control electronics, while high-power actuators are powered separately through an external actuator driver circuit. This separation allows the control node to remain passive-powered while still supporting practical actuator control.

The node also includes edge processing. Sensor data are read and processed inside the microcontroller instead of depending on a cloud platform or external computer for every decision. The node can measure five indoor farming parameters using four sensors: air temperature, relative humidity, light intensity level, growing-medium moisture, and growing medium temperature. Based on received commands and local sensor information, the node can generate control outputs for four actuator channels. This makes the system more practical for indoor farming automation because sensing and control actions can be handled close to the plant zone.

Binary FSK was selected for VLC downlink communication because the system requires reliable low-data-rate command transmission rather than high-speed data transfer. The developed VLC link achieved a data rate of 200 bps with a packet error rate of 0.90 % with distance 1.2 m**,** showing that the proposed optical command link is suitable for short control frames in the developed prototype. Previous VLC research has also shown that frequency-based modulation can improve noise resilience compared with simple OOK systems [11].

Therefore, the main original contribution of this research is the development and validation of a VLC based passive powered control node for indoor farming that integrates RF-free bidirectional optical communication, SLIPT-based solar panel operation, local edge processing, sensor measurement, and actuator control in a single system.

1.5 Objectives

1.5.1 Overall Objective

The overall objective of this research is to design and develop a VLC-based passive-powered control node for indoor farming that can receive optical downlink commands, measure local environmental parameters, generate actuator control signals, transmit feedback through an infrared uplink, and perform local data processing using harvested optical energy.

1.5.2 Specific Objectives

The specific objectives of this research are as follows:

1. To develop a control node for indoor farming that is capable of reading multiple sensors and control multiple actuators.
2. To implement Simultaneous Lightwave Information and Power Transfer (SLIPT) based techniques to supply power to the control node.
3. To establish VLC/IR based data transferring techniques for the control node.
4. To apply edge computing approaches for on-device data processing, decision making and optimized data transmission.

CHAPTER 2. Methodology

2.1 Overview of the Proposed Control Node

Based on the sensing, actuation, communication, and power requirements of indoor farming, this research proposes a VLC-based passive-powered sensing and control module. The overall architecture of the proposed control node is shown in Figure 2.1.

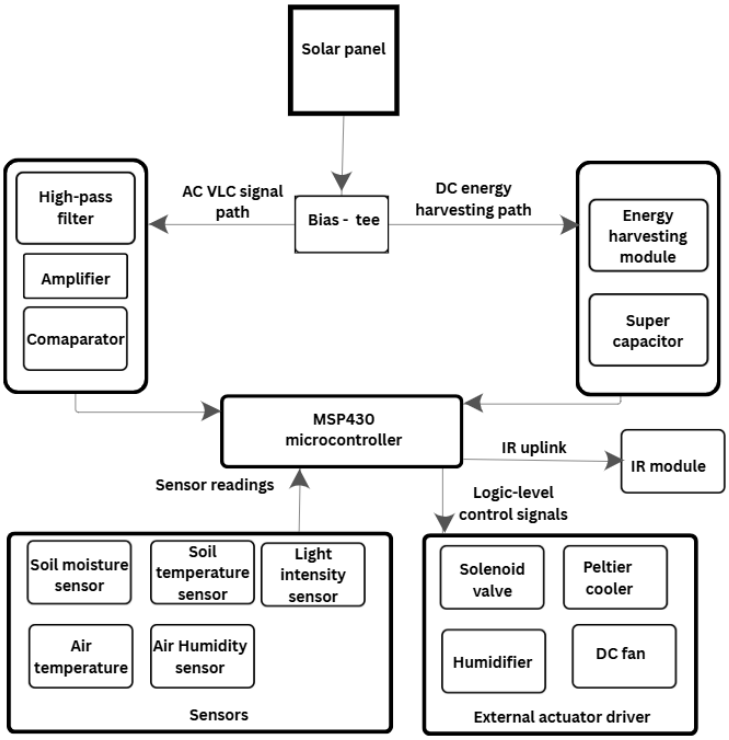


Figure 2.1 : Overall architecture of the proposed VLC-based passive-powered sensing and control node

The system uses a solar panel as a dual-purpose optical interface for energy harvesting and VLC signal reception. The received signal is separated using a bias-tee interface, where the DC component is directed to the energy harvesting path and the AC component is processed through the VLC receiver path.

The VLC receiver path converts the weak AC signal from the solar panel into a digital pulse signal that can be decoded by the MSP430 microcontroller. The power path consists of the energy harvesting module and supercapacitor, which provide the regulated supply required by the low-power electronics of the control node. The MSP430 acts as the central controller and manages VLC frame decoding, sensor reading, actuator control signal generation, and IR uplink communication.

The node reads environmental parameters such as soil moisture, soil temperature, air temperature, relative humidity, and light intensity. Based on received commands and local processing, the MSP430 generates logic level control signals for external actuator driver circuits. The actuator loads are powered separately, while the passive-powered control node only provides the control signals. An IR uplink is used to send acknowledgement and sensor/status data back to the grow light or control system.

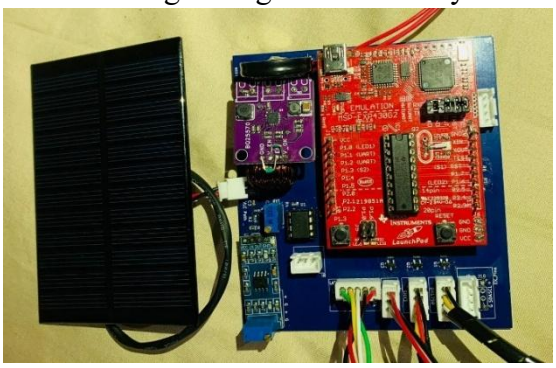


Figure 2.2 : VLC based passive powered control node

This research was carried out in parallel with the development of a VLC enabled grow light system by Group 26. These systems were designed to interface with each other during full system operation. The grow light unit is treated as the external optical transmitter and illumination source required for testing the control node. Some experimental measurements, such as VLC link performance, optical intensity related tests, and system communication validation, were obtained using the common integrated test setup because the transmitter and receiver subsystems operate as an interdependent optical link.

2.2 SLIPT-Based Power and Communication Interface

2.2.1 Overview of SLIPT Concept

Simultaneous Lightwave Information and Power Transfer (SLIPT) is a technique in which light is used for both information transmission and energy transfer. In a SLIPT system, the optical signal received by a photoelectric converter can be used to harvest energy and decode communication data.

In the proposed control node, SLIPT is applied by using a solar panel as the main optical interface. The solar panel receives light from the external modulated grow light unit. The slowly varying DC component of the solar panel output is used for energy harvesting, while the AC component generated by light modulation is extracted for VLC command reception.

Several SLIPT receiver architectures have been reported in the literature. The main methods include time switching, power splitting, space splitting, and signal component separation [4]. Table 2.1 summarizes the main SLIPT receiver architectures reported in the literature and compares their operating principles.

Table 2-1 : Comparison of SLIPT receiver architecture

Method	Description
1.Time switching	This method has low hardware complexity, but energy harvesting and data reception do not occur at the same time.
2.Power splitting	The incoming optical beam is divided into two parts. One part is used for information decoding and the other part is used for energy harvesting. This method requires an optical power-splitting element, which increases mechanical and optical complexity.
3.Space splitting	Different photoelectric converters or different receiving areas are used for energy harvesting and information decoding. This method can support simultaneous operation, but it increases receiver size and hardware cost because separate optical receiving elements are required.
4.Signal component separation	The received electrical signal is separated into DC and AC components. The DC component is directed to the energy harvesting circuit, while the AC component is directed to the information decoding circuit.

In this research, signal component separation was selected using a bias-tee interface. This can be considered as an electrical DC/AC splitting method. It is different from optical power splitting because no optical beam splitter is used. Instead, the electrical output of the solar panel is separated into two paths: a DC energy harvesting path and an AC VLC signal path.

This method was selected because it allows the same solar panel to be used for both energy harvesting and VLC reception, does not require a mechanical switch between harvesting

and decoding, does not require an optical beam splitter or separate photodetector, supports compact hardware implementation, and is suitable for low-data-rate command reception. It also allows the energy harvesting circuit to continue charging while the VLC signal is extracted.

2.2.2 Solar Panel as the Dual-Purpose Optical Interface

In the proposed control node, a small solar panel was used as the main optical interface for both energy harvesting and VLC signal reception. The solar panel converts the received optical energy from the external modulated grow-light unit into an electrical signal. Since the light source contains both an average illumination component and a modulated communication component, the electrical output of the solar panel also contains both DC and AC components.

The DC component of the solar-panel output is produced by the average received light intensity and is used for energy harvesting. This component is directed to the energy-harvesting path, where it charges the supercapacitor through the BQ25570 harvesting module [26]. The AC component is produced by the modulated optical signal and carries the VLC command information. This component is extracted through the AC signal path and processed by the high-pass filter, amplifier module (LM358) [27], comparator (LM393) [28], and MSP430 pulse-counting firmware.

The use of a solar panel provides two main advantages for the proposed passive-powered control node. First, it allows the node to harvest energy from the same optical source used for communication. Second, it reduces the need for a separate photodiode receiver, making the node simpler and more compact.

However, a solar panel has a slower electrical response than a dedicated high-speed photodiode because of its internal capacitance and large junction area. In this research, low-kHz Binary FSK frequencies were used because they could be detected by the solar-panel receiver and processed using a low-power comparator and MSP430 Timer_A pulse-counting method.

The solar-panel output is also affected by transmitter - receiver distance, panel orientation, illumination level, and loading from the energy-harvesting circuit. Therefore, the solar

panel was experimentally characterized, and the receiver front-end was designed to separate the DC harvesting component from the AC VLC signal component.

The response of the solar panel to the modulated optical signal was verified using an oscilloscope. During this test, an external modulated grow light unit was used as the VLC signal source. The transmitter reference signal and the raw electrical output from the solar panel are shown in Figure 2.3.

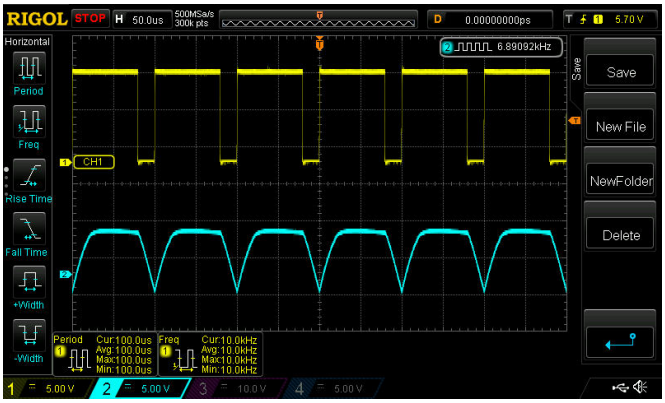


Figure 2.3 : Oscilloscope waveform showing the transmitter reference signal and raw solar-panel VLC output. CH1 -the transmitter signal, and CH2 - the raw solar-panel output.

The waveform shows that the solar panel detects the modulated optical signal and produces a corresponding electrical waveform. However, the raw solar-panel output is not a clean digital signal.

2.2.3 Function of the Bias-Tee Circuit for Solar-Panel-Based SLIPT Interface

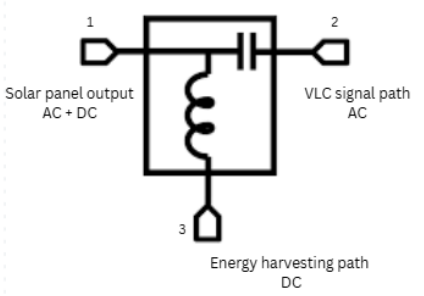


Figure 2.4 : Bias -tee

The solar panel output contains both the harvested energy component and the communication signal component. The DC component is produced by the average light intensity and is useful for charging the energy storage element. The AC component is produced by the modulated light intensity and carries the VLC command information.

Therefore, a bias-tee type circuit was used to separate these two components. The circuit has two main paths:

1. AC – VLC signal path
2. DC – Energy harvesting path

The bias-tee interface used in the proposed control node is shown in Figure 2.4. The inductor provides a low-resistance path for DC energy harvesting while increasing impedance at the VLC modulation frequencies. The capacitor blocks the DC voltage from the solar panel and passes the AC communication signal to the receiver circuit.

The inductor in the DC path was used as a choke between the solar panel and the energy harvesting module. Its purpose was to allow the harvested DC current to flow toward the energy harvesting module while reducing the flow of the AC VLC component into the energy harvesting circuit. This is important because the energy harvesting module and storage capacitor can load the solar panel output and reduce the amplitude of the VLC signal.

The waveform after the coupling capacitor and the waveform after the DC-path inductor are shown in Figure 2.5.

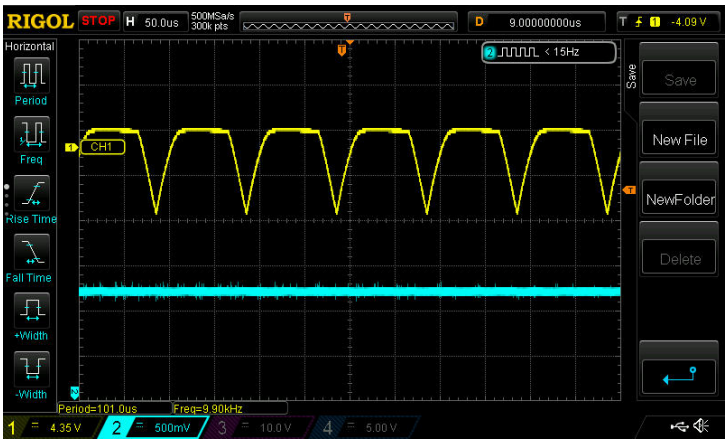


Figure 2.5 : Oscilloscope waveform of the bias-tee output paths. CH1 - yellow shows the AC-coupled VLC signal after the capacitor, and CH2 - blue shows the DC-path signal after the inductor.

This waveform shows that the AC-coupled path contains the modulated VLC signal after the coupling capacitor. In contrast, the waveform after the inductor shows significantly

reduced AC variation, indicating that the inductor helps to limit the flow of the VLC AC component into the energy-harvesting path.

The inductive reactance is given by:

$$X_L = 2\pi f L \quad \ldots\ldots\ldots\ldots (1)$$

Where;

X_L - inductive reactance

f - modulation frequency

L - inductance.

According to this equation, a higher inductance gives higher impedance at the VLC frequencies.

However, the inductor was not selected only from the theoretical reactance calculation. Practical inductors contain parasitic resistance, parasitic capacitance, magnetic pickup, and possible resonance effects with the solar panel capacitance and the input capacitance of the energy harvesting module. Therefore, the inductor value and construction were finalized experimentally. A hand-wound inductor was made using 0.53 mm copper wire with 100 turns to get 140 mH inductance. The thicker copper wire was selected to reduce DC resistance so that the harvested current could pass to the energy harvesting module with minimum voltage drop. The number of turns was selected experimentally to obtain sufficient impedance at the Binary FSK frequencies while avoiding excessive noise in the DC path.

In the bias-tee interface, the capacitor in the AC path was used to couple the modulated VLC signal from the solar panel to the receiver circuit while blocking the DC voltage generated by the solar panel. If the capacitance is too small, its impedance at the selected VLC frequencies becomes high, and the received signal amplitude at the comparator input becomes weak.

Impedance of a capacitor is given by:

$$Xc = \frac{1}{2\pi fc} \quad \ldots\ldots\ldots\ldots (2)$$

A higher capacitance gives lower impedance at the VLC frequencies.

The effect of the DC-path inductor was verified using an oscilloscope by comparing the solar-panel output waveform with the waveform observed after the inductor in the energy harvesting path. The measured waveforms are shown in Figure 2.6.

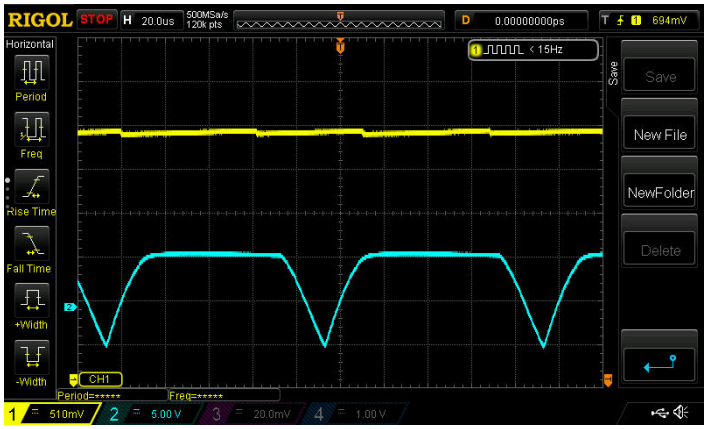


Figure 2.6: Oscilloscope comparison of the solar-panel output waveform and the DC-path output after the inductor. CH1 - DC-path output after the inductor, and CH2 - solar-panel VLC waveform.

The oscilloscope observation shows that the solar-panel output contains the modulated VLC signal, while the waveform after the inductor has reduced AC variation.

2.3 High pass filter

After separating the AC VLC signal from the solar panel output, the signal was passed through a high-pass filtering and comparator biasing stage.

The high-pass filter was mainly used to reduce ambient light noise and low-frequency illumination variations. In indoor environments, artificial lighting may introduce mains related flicker components around 50 Hz/60Hz, ripple components around 100 Hz may also appear. These low frequency components are unwanted for VLC reception because they can cause false triggering at the comparator input.

The filter cutoff frequency had to be high enough to suppress 50 Hz ambient light flicker, but low enough to pass the 3 kHz and 5 kHz VLC signals without significant attenuation. Selecting a cutoff close to 50 Hz would not provide enough rejection of flicker and slow illumination changes. A cutoff near 150 Hz gives better low-frequency noise rejection while still remaining much lower than the selected VLC frequencies.

The cutoff frequency of the first-order high-pass filter is given by:

$$F_c = \frac{1}{2\pi RC} \quad \text{…….. (3)}$$

R = 10 KΩ

C = 100 nF

Therefore,

$F_c \approx 159$ Hz

The signal amplitude at the cutoff frequency is approximately 70.7% of the passband amplitude. Therefore, frequencies below the cutoff are attenuated more strongly, while frequencies above the cutoff pass with less attenuation. Since both VLC frequencies are well above the -3 dB cutoff frequency, they remain inside the effective passband of the receiver front-end.

2.4 Comparator and Biasing Components

After AC coupling and high-pass filtering, the DC level from the solar panel is removed. As a result, the received VLC signal becomes a small AC waveform. Since the comparator operates from a single 3.3 V supply, its input must remain within the valid input voltage range and must have a defined reference level.

Therefore, biasing components were added to set a stable operating point for the comparator input after amplifying the signal using amplifier module. This allows the comparator to compare the incoming waveform against a defined threshold and convert the weak solar-panel signal into a clean digital pulse train.

The biasing stage is important because the amplitude of the solar-panel VLC signal changes with the received optical power, distance from the transmitter, solar panel alignment, ambient lighting condition, and loading effect of the energy harvesting path. A fixed zero-volt reference is not suitable because the AC-coupled waveform may not remain within the comparator's valid input range. By using a bias network, the comparator threshold can be adjusted to obtain stable detection during practical testing.

The comparator output is then connected to the MSP430 input pin. Instead of directly measuring analog solar panel waveform, the MSP430 receives the digital waveform generated by the comparator and measures its frequency to identify the Binary FSK symbol. Therefore, the purpose of the analog front-end is to produce a stable digital waveform from a weak and noisy solar-panel VLC signal. This approach is consistent with solar-panel-based VLC receiver designs where the analog front-end filters and digitizes the received signal before microcontroller processing.

The transmitter signal, the solar-panel output, and the comparator output waveforms are shown in Figure 2.7. The solar panel converts the modulated optical signal into an electrical waveform, and the comparator converts the conditioned analog signal into a digital pulse waveform suitable for the MSP430 Timer_A pulse-counting input.

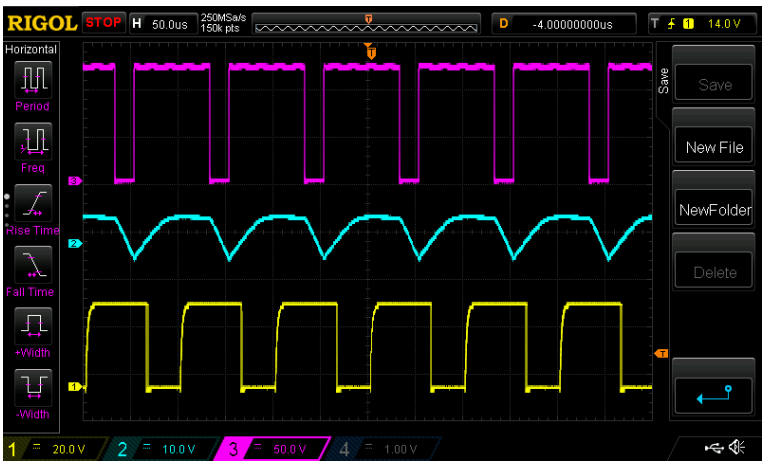


Figure 2.7 : Oscilloscope verification of the VLC signal path. CH1-shows the comparator digital output, CH2- shows the solar-panel output, and CH3- shows the transmitter signal

The solar-panel output is not a clean digital signal. After biasing, amplification, and comparison, the comparator output becomes a clean digital pulse waveform. This digital waveform can be counted by the MSP430 Timer_A peripheral to decode the Binary FSK symbols.

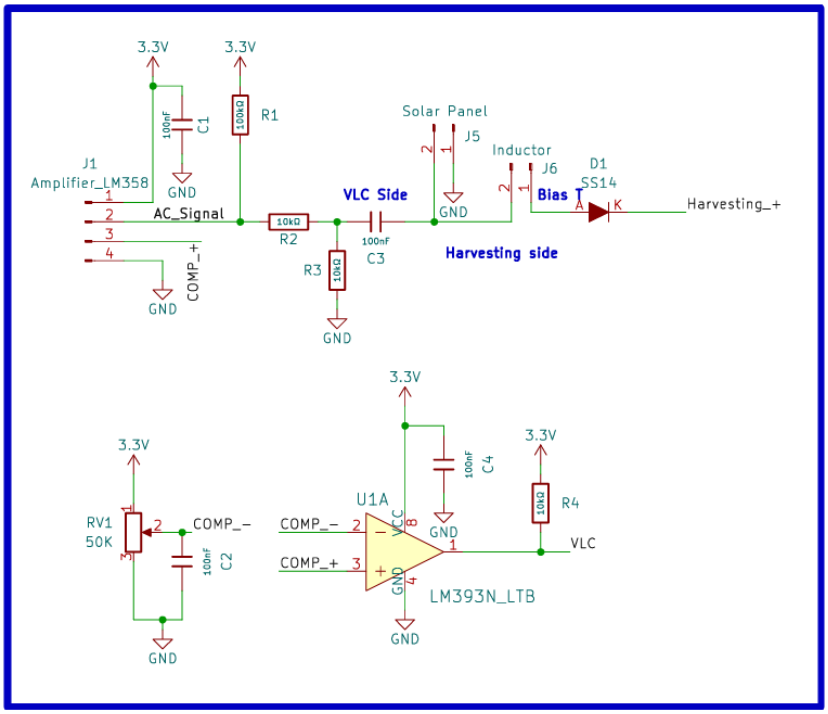


Figure 2.8 : VLC Analog-front end schematic diagram

2.5 VLC Downlink and Binary FSK Frame Decoding

2.5.1 Overview of VLC Modulation Methods

Visible Light Communication transmits information by varying the optical output of an LED light source. Several modulation methods can be used in VLC systems, and the selection depends on the required data rate, receiver type, circuit complexity, noise environment, and illumination requirements. In this research, the receiver is based on a solar panel rather than a high-speed photodiode. Therefore, the selected modulation method had to be simple, low frequency, suitable for a solar panel receiver, and reliable for short command frames.

Common VLC modulation techniques include:

1. On-Off Keying (OOK)
2. Amplitude Shift Keying (ASK)
3. Pulse Width Modulation (PWM)

4. Pulse Position Modulation (PPM)
5. Variable Pulse Position Modulation (VPPM)
6. Orthogonal Frequency Division Multiplexing (OFDM)
7. chirp-based modulation
8. Frequency Shift Keying (FSK)

Since indoor farming environments may contain ambient light variations, reflections, and solar-panel receiver limitations, modulation selection is an important part of the receiver design.

Table 2-2 : Comparison of VLC modulation methods considered for the proposed control node

VLC modulation method	Description
ON – OFF keying (OOK)	OOK is the simplest VLC modulation method. In OOK, the LED is turned ON to transmit one logic level and OFF or reduced to transmit the other logic level. It depends strongly on received signal amplitude. In a solar panel-based receiver, the received amplitude changes with distance, alignment, light intensity, and harvesting load. For this reason, OOK was not selected as the final method for the proposed control node.
Amplitude shift keying (ASK)	ASK changes the optical intensity level according to the transmitted data. It can transmit data without fully switching the light OFF, but the receiver must still detect amplitude changes accurately. Therefore, it is affected by distance, solar-panel orientation, and ambient light variation.
PWM or duty-cycle-based modulation	PWM changes the duty cycle of the LED signal to represent data. This is compatible with LED dimming, but changing duty cycle also changes the average optical power. In indoor farming, large changes in average optical output are not preferred because the

	light source also supports plant illumination and energy harvesting.
OFDM and other high-data-rate schemes	OFDM can achieve high throughput in VLC systems, but it requires more complex signal generation, analog-to-digital conversion, and signal processing. This complexity is not suitable for the proposed low power control node application.
Binary Frequency Shift Keying (Binary FSK)	Binary FSK represents binary data using two different modulation frequencies. The LED remains active and the transmitter changes only the modulation frequency. The receiver can decode data by counting pulses within a fixed sampling window. This is suitable for the MSP430 because Timer_A can count comparator output pulses without continuous ADC sampling. Therefore, Binary FSK was selected as the final VLC modulation method.

Binary FSK was selected because the proposed control node requires reliable low-data-rate command transmission rather than high-speed data transfer. Previous VLC research has also shown that frequency-based modulation can improve noise resilience compared with simple OOK systems [4].

2.5.2 VLC Downlink implementation

The VLC downlink was implemented using Binary Frequency Shift Keying. In this method, two optical modulation frequencies represent the binary data symbols. Instead of switching the light fully ON and OFF for each bit, the transmitter changes the PWM frequency of the LED output while maintaining a constant duty cycle. This method is suitable for the proposed control node because the solar-panel receiver output amplitude can vary with distance, alignment, ambient light, and energy harvesting load. Frequency-

based detection is therefore more suitable than relying only on amplitude threshold detection.

For testing the VLC receiver of the control node, an external modulated LED grow-light unit was used as the VLC downlink transmitter. The transmitter generated Binary FSK optical signals using the selected frequencies for idle, logic 0, and logic 1 states.

During the VLC downlink transmission, the external transmitter modulated the white, red, and blue LED channels together using PWM frequency variation. The duty cycle was kept constant during transmission so that data were represented mainly by frequency changes rather than large average-intensity changes. This allowed the solar-panel receiver to detect Binary FSK symbols while maintaining continuous illumination from the light source.

Table 2-3 : Selected frequencies for VLC downlink

State	Frequency
Idle	10 kHz
Binary 0	3 kHz
Binary 1	5 kHz

An idle frequency of 10 kHz was used when no data frame was being transmitted. This kept the light source continuously modulated and allowed the receiver to identify a stable and distinguishable idle condition. The data frequencies of 3 kHz and 5 kHz were selected after practical frequency-response and PER tests. Earlier tests used lower-frequency pairs such as 800 Hz/1000 Hz and 1.5 kHz/2 kHz. These low frequency configurations were detectable by the solar-panel receiver, but the final 3 kHz and 5 kHz pair provided better frequency separation for pulse count-based decoding and supported a higher data rate.

The data frequencies of 3 kHz and 5 kHz were selected because they are much higher than the high-pass filter cutoff frequency of approximately 150 Hz and much higher than indoor 50 Hz lighting flicker. Therefore, low frequency ambient light noise is attenuated while the useful VLC frequencies pass through the receiver front-end with very low attenuation. These frequencies are also low enough to be detected reliably by the solar-panel receiver,

analog front-end, comparator and MSP430 pulse-counting method. Solar panels have limited bandwidth because of their internal capacitance and low-pass behavior [1]. Therefore, very high VLC frequencies were avoided.

Three data rates were tested during development: 50 bps, 100 bps, 200 bps. These correspond to symbol durations of 20 ms, 10 ms, and 5 ms respectively. However, 200 bps was selected as the final operating data rate because it provided a better balance between transmission time and decoding reliability. The proposed VLC downlink transmits short command frames rather than continuous high-volume data. Therefore, reducing frame transmission time is useful, provided that packet reliability remains acceptable.

The symbol duration used in the final VLC transmitter code was 5 ms. Each symbol represents one binary bit. Therefore, the raw data rate is:

$$Data\ rate = \frac{1}{Symbol\ duration} \quad \text{..... (4)}$$

$$= \frac{1}{5 \times 10^{-3}}$$

= 200 bps

2.5.3 VLC frame structure

The VLC downlink uses a custom frame-based protocol implemented over a Binary FSK physical layer. The purpose of the frame structure is to allow the MSP430 receiver to detect the beginning of a transmission, synchronize to the received bitstream, extract the command bytes, and validate the received packet before accepting it as a valid control command.

Before transmitting a command frame, the grow light unit transmitter keeps the LED output in the idle state (10 kHz). For low power receiver operation, a wake burst is transmitted before the first VLC frame of a transaction. This wake burst allows the MSP430 to detect the presence of an incoming optical command while operating with a timer-based wake scan method. After the wake burst and idle period, the actual VLC frame is transmitted using the selected Binary FSK symbols.

The frame structure used in the final implementation is shown in Figure 2.9.



Figure 2.9 : VLC downlink frame structure used for Binary FSK command transmission

The frame starts with 20 start symbols. These start symbols are transmitted as logic 0 symbols. Since each symbol duration is 5 ms, the total start sequence duration is 100 ms. The start burst is transmitted using the logic 0 frequency of 3.0 kHz. After the start burst, an 8bit synchronization byte is transmitted. The synchronization byte used in this work is 0xA5, which helps the MSP430 receiver align with the incoming bitstream before reading the data bytes.

After the start symbols, an 8-bit synchronization byte is transmitted. The synchronization byte used in this work is 0xA5. This byte helps the MSP430 receiver align with the incoming bitstream before reading the data bytes.

After synchronization, the address byte is transmitted. The address byte identifies the intended receiver and is formed using the grow light ID and control node ID. The control byte contains the command class and sequence number. In the implemented frame structure, the upper three bits of the control byte represent the command class, while the lower five bits represent the sequence number. The data byte carries the actuator, sensor or command related value. Finally, a CRC byte is transmitted for error detection.

The main frame fields are summarized in Table 2.4.

Table 2-4 :VLC frame fields summary

Field	Length	Description
Start symbols	20 symbols	100 ms start burst transmitted at the 3.0 kHz logic 0 frequency
Sync byte	8 bits	Synchronization byte, 0xA5
Address byte	8 bits	Identifies the intended grow-light and control-node address

Control byte	8 bits	Defines command type or sequence value during testing
Target byte	8 bits	Carries actuator, sensor, or command-related value
CRC byte	8 bits	CRC-8 value calculated from address, control, and target bytes
Tail symbols	2 bits	Logic 0 symbols used after the frame before returning to idle

The CRC-8 method used in the transmitter and receiver uses polynomial 0x07 with an initial value of 0x00. The CRC is calculated using the address, control, and target bytes. At the receiver side, the MSP430 reconstructs the received bytes and calculates the CRC again. The received frame is accepted only if the calculated CRC matches the received CRC byte. If the CRC does not match, the frame is rejected. This prevents corrupted VLC packets from being used for actuator control.

2.5.4 MSP430 Receiver Decoding Method

The VLC receiver converts the solar-panel AC signal into a digital pulse signal using the receiver front-end and comparator. The comparator output is connected to P1.0 of the MSP430, which is configured as the Timer_A external clock input. Therefore, the MSP430 does not continuously sample the analog waveform using the ADC. Instead, Timer_A counts the incoming comparator pulses during a fixed sampling window.

The final receiver code uses a 5 ms symbol duration. Each symbol is sampled using a centered timing window consisting of a 0.5 ms pre-guard interval, a 4 ms pulse-count interval, and a 0.5 ms post-guard interval. The guard intervals reduce the effect of symbol-boundary timing errors, while the 4 ms count window is used to classify the received frequency.

For the selected frequencies, the expected pulse counts in the 4 ms count window are approximately shows in Table 2.5.

Table 2-5 : Expected edge counts for each frequency

Received state	Frequency	Expected count in 6 ms
Binary 0	3 kHz	12 pulses

Binary 1	5 kHz	20 pulses
Idle	10 kHz	40 pulses

The receiver uses threshold-based symbol classification. A count greater than or equal to the idle threshold, the symbol is classified as idle. For data symbols, a count below 16 is classified as logic 0, while a count equal to or greater than 16 is classified as logic 1. This allows the MSP430 to decode Binary FSK symbols using pulse counting instead of amplitude measurement.

The receiver does not assume that a predefined frame has been received. Instead, it detects the optical signal, searches for the start condition, reads the synchronization byte, reconstructs the address, control, data, and CRC bytes, and validates the frame using CRC-8. Only frames with valid synchronization, correct address, and matching CRC are treated as valid received packets. The packet error rate testing method was used to compare the reliability of different data-rates and to verify the final 200 bps Binary FSK VLC downlink.

2.5.5 Low-Power VLC Wake-Scan Method

Since the control node is powered from harvested energy, the receiver firmware was designed to reduce continuous active operation. The MSP430 does not use a P1.0 edge interrupt for wake-up because the 10 kHz idle signal would create continuous edges and keep the microcontroller active. Instead, the receiver uses a timer-based wake-scan method.

In this method, the MSP430 remains in a low power state for most of the time and wakes periodically using the watchdog timer interval. After each timer wake-up, the firmware briefly configures P1.0 as the Timer_A external clock input and samples the incoming pulse count. The transmitter sends a repeated 5 kHz wake burst before the first VLC frame of a transaction. If the MSP430 detects the expected wake-burst frequency during the scan window, it remains active and waits for the actual VLC frame. If the wake condition is not detected, the MSP430 returns to low-power operation.

This method reduces unnecessary active time while still allowing the control node to detect incoming VLC commands from the grow light transmitter unit. This approach avoids continuous ADC sampling and avoids continuous CPU activity during the idle state.

2.5.6 Frequency Response Test and FSK Frequency Selection

A frequency response test was carried out to select suitable Binary FSK frequencies for the VLC downlink. Since the solar panel was used as the optical receiver, it was necessary to evaluate how the received AC signal changed with modulation frequency and transmitter - receiver distance. The solar panel and the receiver front-end do not respond equally to all modulation frequencies, therefore this test is useful to select correct frequency ranges.

During the test, the LED transmitter was driven using different square-wave modulation frequencies while maintaining the same duty cycle (80%). The solar panel was placed at different distances from the grow light unit and the received AC signal level was measured as peak-to-peak voltage. The test was repeated for several frequencies in the low kHz range and extended to higher frequencies to observe the reduction in received signal level. Candidate Binary FSK frequency pairs were then tested by observing the received signal amplitude, comparator output stability, and digital edge-count behavior at the receiver.

The purpose of this experiment was to identify a frequency pair that produced a detectable and stable received signal while maintaining enough separation between logic 0 and logic 1. The selected frequency pair also had to be suitable for low data-rate command transmission using the MSP430 microcontroller.

The frequency response test was followed by data rate testing. The VLC downlink was evaluated at 50 bps, 100 bps, and 200 bps using the selected higher-frequency FSK range. These data rates correspond to symbol durations of 20ms, 10ms, and 5ms respectively. The final implementation used a 5ms symbol duration, giving a raw data rate of 200 bps.

The 200bps configuration was selected because the proposed VLC downlink transmits short command frames rather than continuous high-volume data. Therefore, reducing the command frame transmission time is useful as long as the packet error rate remains acceptable. Therefore, the final VLC downlink used 3 kHz for logic 0, 5 kHz for logic 1, 10 kHz for idle, and a symbol duration of 5ms.

2.6 Energy Harvesting Path and Power Regulation

The DC path of the SLIPT interface was used to harvest optical energy from the solar panel and supply power to the low power electronics of the control node. In the final control node, a hand-wound inductor with a measured inductance of 140.53 mH was used in the DC path to reduce AC signal leakage toward the harvesting module while allowing the harvested DC current to flow to the energy harvesting path. The energy harvesting module then extracts energy from the solar panel and stores it in the supercapacitor. The stored energy is used to power the MSP430 microcontroller, sensor interface circuits, VLC receiver circuit, and IR uplink circuit.

2.6.1 Solar Panel Characterization

The solar panel used in the control node did not have a complete manufacturer datasheet. Therefore, experimental characterization was carried out to understand its electrical behavior under the grow light unit. This characterization was important because the same solar panel is used as both the energy harvesting element and the VLC receiver.

The voltage - current characteristic of the solar panel was obtained using a variable resistive load method. The solar panel was placed under a fixed illumination condition, and different load resistance values were connected across the panel terminals. For each load condition, the terminal voltage was measured, and the corresponding load current was obtained. The output power was then calculated using:

$$P = V \times I \quad \ldots\ldots\ldots\ldots (5)$$

Where;

P = output power (W)

V = measured terminal voltage (V)

I = measured load current (A)

The open - circuit voltage was measured when no load was connected, and the short-circuit current was measured when the panel terminals were connected through an ammeter. The purpose of this test was to estimate solar panel's practical operating behavior under the experimental lighting condition used in this research. The obtained curves were used to identify the approximate open-circuit voltage, short-circuit current, maximum power point (MPP), and the available power range for the energy harvesting path.

2.6.2 Energy Harvesting Module

A photovoltaic energy harvesting module (BQ25570) was used in the control node. The BQ25570 is suitable for low power photovoltaic harvesting because it is designed to extract energy from low-power, high-impedance sources such as photovoltaic cells [29]. Previous indoor PV energy harvesting research has evaluated commercial FOCV-based PMUs, including the BQ25570, for low-area photovoltaic cells under indoor illumination [30] . That study also explains that FOCV-based power management units (PMUs) regulate the operating voltage of the PV cell around its maximum power point (MPP) by periodically sampling the open-circuit voltage [30].

In the proposed control node, the harvesting module has three main functions.

1. It receives the DC energy from the solar panel through the DC path of the bias tee.
2. It stores the harvested energy in a supercapacitor, which acts as an energy source.
3. It provides a regulated output voltage (3.3 V) to the low power electronics when sufficient energy is available.

This energy harvested operation is important because the available light energy can vary with illumination level, distance from the light source, and solar panel orientation [30].

A supercapacitor was used as the energy storage element instead of a conventional rechargeable battery. That supercapacitor stores the harvested energy and supplies current during active operation of the node. This is suitable for the proposed system because the control node operates in low power cycles, where the harvested energy is accumulated over time and then used during sensing, communication, or control activity.

The stored energy in the supercapacitor can be expressed as:

$$E = \frac{1}{2} CV^2 \quad \text{..... (6)}$$

where E is the stored energy, C is the capacitance, and V is the capacitor voltage.

The supercapacitor acts as the temporary energy source for the control node. When the harvested energy raises the stored voltage above the required operating level, the regulated output becomes available to the control electronics. When the stored voltage falls below the required level, the node cannot continue active operation and must wait until the capacitor is recharged.

The control node requires a minimum operating voltage. Therefore, the usable energy between the upper and lower operating voltages can be expressed as:

$$E = \frac{1}{2} C \left(V_{high}{}^2 - V_{low}{}^2 \right) \quad \text{..... (7)}$$

Where:

V_{high} – maximum capacitor charging voltage

V_{low} – minimum usable voltage (3.3 V)

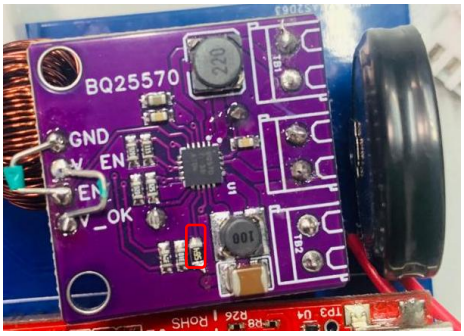


Figure 2.10 : Feedback resistor modification on the BQ25570 module for 3.3 V regulated output

The BQ25570 module initially provided an output voltage of 2.5 V. However, the control node electronics, including the MSP430, sensors, comparator, amplifier module and IR module, were designed to operate from a 3.3 V logic level. Therefore, the output voltage of the harvesting module was changed from 2.5 V to 3.3 V by modifying the output voltage feedback resistor network [26]. After the modification, the output voltage was measured and verified as approximately 3.3 V before connecting the control-node electronics.

The regulated 3.3 V output from the harvesting module was used only for the low power electronics of the control node. The actuator loads were not powered from the harvested energy because devices such as fans, pumps, solenoid valves, humidifiers, and cooling modules require significantly higher current than the solar panel harvesting path can provide. Therefore, the actuator system was separated into two parts. The passive-powered control node generates only the logic level actuator control signals. These signals are sent to an external actuator driver circuit, where a separate actuator power supply provides the required load current. This separation prevents the harvested energy system from being overloaded and allows the control node to remain passive-powered.

2.6.3 MSP430 Based Low Power Control Unit

The MSP430 microcontroller was used as the central low power controller of the control node. The microcontroller was selected because the control node is powered from harvested optical energy from the grow light, and therefore the processing unit must operate with low active current, support sleep mode operation, and provide simple hardware peripherals for sensing and communication. The MSP430G2 provides low power operating modes, timers, digital I/O, ADC functionality, and serial communication interfaces, which are suitable for the proposed sensing and control node [31].

The selection of the MSP430 is also supported by previous low power optical communication and energy harvesting node designs. In a greenhouse VLC sensing system, MSP430 microcontroller-based sensor tags were proposed together with a solar panel, energy harvester, comparator, and filtering stages for low-power VLC reception [4]. Similarly, Retro-VLC research used an MSP430 microcontroller in a battery-free optical tag, where the MCU remained in sleep mode for most of the time and woke only for short timing and decoding tasks [6].

In the developed system, the MSP430 performs four main functions:

- ✓ VLC frame decoding
- ✓ sensor data acquisition
- ✓ actuator control signal generation
- ✓ IR uplink communication

The output of the VLC analog front-end is connected to the MSP430 as a digital pulse signal. Since the solar panel signal is already filtered and converted into a digital waveform by the comparator, the microcontroller does not need to continuously sample the analog waveform using the ADC. Instead, the Timer_A peripheral is used to count the number of pulses within a fixed symbol period. Based on the pulse count, the firmware identifies whether the received Binary FSK symbol represents binary 0, binary 1, or the idle state. This pulse counting method reduces computational complexity and supports low power operation.

The MSP430 also controls the sensor acquisition sequence. The soil moisture sensor is read through an ADC input, while the digital sensors are accessed through the I2C interface. To reduce unnecessary power consumption, selected sensors are powered only during the measurement period using MOSFET switching.

The actuator outputs are generated only as low voltage logic control signals from the MSP430. The MSP430 sends control signals to an external actuator driver circuit, where the actuator loads are powered by a separate supply.

Low power firmware operation is a key requirement of the control node. The MSP430 remains in a low power sleep mode state when sensing, decoding, or communication tasks are not required. During an active cycle, the microcontroller wakes up, enables the required sensor circuits, reads the sensor values, processes the data, updates the actuator control signals, and sends acknowledgement or status information through the IR uplink when required. After completing these tasks, the microcontroller returns to low power state.

The MSP430 supports different low power modes that allow the microcontroller, when full active operation is not required [31].

Table 2-6 : MSP430 low-power modes considered for the control node

Mode	**CPU and clock status**
LPM0	CPU off, MCLK OFF, SMCLK and ACLK active
LPM1	CPU off, MCLK OFF, SMCLK and ACLK active; DCO can be disabled if not required
LPM2	CPU off, MCLK and SMCLK OFF, ACLK active

LPM3	CPU off, MCLK, SMCLK, and DCO OFF; ACLK active
LPM4	CPU off, All clocks OFF

The low-power modes considered for the control node are summarized in Table 2.6. In this control node, LPM3 was selected as the main sleep mode. During LPM3 operation, the CPU and high-speed clocks are disabled, while ACLK remains active. This allows the Timer_A module to continue operating from the low frequency clock and wake the microcontroller after the required sleep period. Therefore, the node can remain in a low-power state between sensing and control cycles without losing the ability to wake up periodically.

The average current consumption of the microcontroller-based system depends on the active time and sleep time of the firmware cycle. This can be represented as:

$$I_{avg} = \frac{I_{active} \times t_{active} + I_{sleep} \times t_{sleep}}{t_{active} + t_{sleep}} \quad \ldots\ldots\ldots\ldots (8)$$

Where;

I_{avg} – average current consumption

I_{active} – current during sensing, decoding and communication

I_{sleep} – current during low power mode

t_{active} – active time

t_{sleep} – sleep time

This relationship shows that reducing active time and increasing sleep duration are important for passive-powered operation. Overall, the MSP430 acts as the main embedded controller that connects the SLIPT-based power system, solar-panel VLC receiver, sensor interface, actuator control interface, and IR uplink. Its low power operation and timer-based pulse counting method make it suitable for the developed VLC-based passive-powered control node.

2.7 Sensor Selection Basis for Strawberry - Based Environmental Control

The sensor subsystem was designed to measure the main environmental parameters required for strawberry based controlled environment cultivation. The selected parameters were air temperature, relative humidity, growing-medium moisture, growing-medium temperature, and light intensity. These parameters were selected because they are directly related to air condition regulation, irrigation control, root zone monitoring and grow light unit evaluation.

Four sensors were integrated with the MSP430-based control node. The SHT3X sensor was used to measure air temperature and relative humidity near the plant canopy. The capacitive soil moisture sensor was used to estimate the moisture condition of the growing medium. The DS18B20 digital temperature sensor was used to measure growing medium temperature near the root zone. The BH1750FVI light sensor was used to measure the light intensity reaching the plant area. Together, these sensors allowed the control node to observe the canopy air condition, root zone condition, growing-medium moisture and grow light condition.

The selected sensors were chosen based on their suitability for low power embedded operation and compatibility with the MSP430 microcontroller.

The SHT3X sensor was placed near the canopy level to measure the air temperature and relative humidity around the leaves, flowers, and fruit. This position was selected because canopy level air condition is more relevant to plant response than air temperature measured far away from the plant. The SHT3X was suitable for this task because it provides calibrated digital temperature and humidity readings through an I2C interface [32].

The capacitive soil moisture sensor [33] was inserted into the growing medium near the root zone to estimate the water condition available to the plant. The sensor was not placed directly below the irrigation inlet or near the tray wall because those positions could give moisture readings that do not represent the overall root zone condition.

The DS18B20 sensor was inserted into the growing medium near the active root zone area to monitor growing medium temperature. This measurement was included because the root zone temperature can differ from the surrounding air temperature. The sensor was not pressed directly against the plant crown or roots. The DS18B20 was suitable for this task because it is a digital temperature sensor with a 1-Wire interface [34].

The BH1750FVI sensor was placed near canopy height and directed toward the grow light to measure the light level reaching the plant area. The sensor was not placed under leaves, behind the control-node body, near tray edges, or below support structures because these positions could produce shadowed or unrepresentative readings. The BH1750FVI measures illuminance in lux. Although it does not directly measure PPFD, it provides a practical digital indication of relative light-intensity variation inside the growing area. The sensor was suitable for the control node because it provides digital light intensity data through an I2C interface [35].

During operation, the MSP430 reads the sensor values after a valid VLC command or sensor request is received. The measured values are then processed and compared with the selected strawberry threshold values in the firmware. Based on these comparisons, the control node generates logic level actuator control signals for functions such as ventilation, humidification, irrigation, and cooling. The measured sensor values and actuator states are then sent back to the grow light unit through the IR uplink. This sensor arrangement allows the proposed control node to operate as a local environmental monitoring and control unit for strawberry-based controlled cultivation.

2.8 Calibration and Filtering Application

2.8.1 Capacitive Soil Moisture Sensor Calibration and Filtering

The capacitive soil moisture sensor was used to measure the relative moisture condition around the plant root zone. The sensor produced an analogue voltage output, and this output was connected to the ADC input of the MSP430 microcontroller. The MSP430G2553 uses a 10-bit ADC module, so the analogue sensor output was converted into a digital ADC value in the range of 0 to 1023 [31].

The sensor was calibrated using two reference conditions. The dry reference value was measured when the sensor probe was kept in normal air without soil or water. This condition was taken as 0% relative moisture. The wet reference value was measured when the sensor probe was placed in water. This condition was taken as 100% relative moisture. From the experimental measurements, the dry reference ADC value was 820 and the wet reference ADC value was 560.

$$Relative\ moisture\ (\%) = \frac{Dry\ reference\ ADC - Measured\ ADC}{Dry\ reference\ ADC - Wet\ reference\ ADC} \times 100 \ldots\ldots\ldots (9)$$

$$Relative\ moisture\ (\%) = \frac{820 - Measured\ ADC}{820 - 560} \times 100 \ldots\ldots\ldots\ldots (10)$$

The capacitive sensor output decreased when the moisture level increased. Therefore, the dry reference value was higher than the wet reference value. After calculating the relative moisture percentage, the value was limited to the range of 0% to 100%. If the calculated value was below 0%, it was set to 0%. If the calculated value was above 100%, it was set to 100%. This prevented invalid values from entering the actuator control logic.

For filtering, two filtering stages were applied to the soil moisture reading [36]. The first stage was applied to the raw ADC readings and the second stage was applied after converting the ADC value into relative moisture percentage. After powering the soil moisture sensor, the code first discarded five ADC readings. These readings were ignored because the analogue output can be unstable immediately after power-up. After that, 10 ADC readings were collected. From these ten readings, the highest ADC value and the lowest ADC value were removed. The remaining eight readings were averaged to obtain a stable ADC value. This method reduced the effect of sudden noise or one abnormal ADC reading. The trimmed ADC average was then used in the calibration equation to calculate the relative moisture percentage. After calibration, an exponential moving average filter was applied to the moisture percentage. The EMA filter was used to smooth the final moisture value between measurement cycles.

The filtered moisture value is calculated using:

$$Filtered\ Moisture = previous\ filtered\ moisture + \alpha \times (New\ moisture - previous\ filtered\ moisture) \quad \ldots\ldots\ldots\ldots (11)$$

In this programmer, α was selected as 0.2. Therefore, the new moisture reading contributed 20% to the updated value, while the previous filtered value contributed 80%. This reduced sudden changes in the final moisture value and made the irrigation decision more stable. During the first measurement cycle, there was no previous filtered value. Therefore, the first calibrated moisture value was used directly as the initial filtered value. From the second cycle onward, the EMA equation was applied. The final filtered moisture value was used for irrigation control and was included in the IR feedback packet.

2.8.2 SHT3X Air Temperature and Humidity Calibration and Filtering

The SHT3X sensor was used to measure air temperature and relative humidity inside the controlled growing environment. The sensor communicates with the MSP430 microcontroller through the I²C interface and provides factory-calibrated digital output values for both parameters [32].

The raw temperature and humidity counts received from the sensor were converted into engineering units using the conversion equations specified in the SHT3X datasheet [32].

The temperature conversion:

$$\text{Temperature (°C)} = -45 + \frac{175 \times \text{Raw Temperature Count}}{65535} \quad \ldots\ldots\ldots\ldots (12)$$

The relative humidity conversion:

$$\text{Relative Humidity (\%RH)} = \frac{100 \times \text{Raw Humidity Count}}{65535} \quad \ldots\ldots\ldots\ldots (13)$$

These equations transformed the 16-bit digital outputs into temperature in degrees Celsius and relative humidity in percentage. Since the SHT3X sensor is factory calibrated by the manufacturer, no additional calibration offset was applied in the current implementation [32]. To improve measurement stability, filtering was applied after data conversion. The first sensor reading obtained after power-up was discarded because

transient effects may occur immediately after sensor initialization. Five valid measurements were then collected. For both temperature and humidity, the highest and lowest readings were removed, and the remaining three readings were averaged. This trimmed-mean approach reduced the influence of occasional abnormal measurements and communication disturbances. After the trimmed-mean calculation, an EMA filter was applied separately to temperature and humidity values. The EMA filter reduced short-term fluctuations while preserving the ability to track gradual environmental changes.

2.8.3 DS18B20 Growing-Medium Temperature Calibration and Filtering

The DS18B20 sensor was used to measure the growing-medium temperature near the plant root zone. It is a digital temperature sensor that communicates using the 1-Wire interface. The DS18B20 supports 9-bit to 12-bit temperature resolution, and at 12-bit resolution each raw count represents 0.0625 °C [34]. The raw temperature value was converted into degrees Celsius using the datasheet-based equation:

$$\text{Temperature (°C)} = \frac{\text{Raw Temperature Count}}{16} \qquad \text{…….. (14)}$$

Since the DS18B20 is a digital temperature sensor, dry and wet calibration was not required. The calibration offset was set to zero, so the converted value was used as the calibrated growing medium temperature. Before accepting a sample, the code checked whether the sensor was detected on the 1-Wire bus. The scratchpad CRC was also checked. If the sensor was not detected or the CRC check failed, that sample was rejected. For filtering, five DS18B20 samples were attempted in each measurement cycle. Only valid samples were averaged.

$$\text{Average Temperature} = \frac{\text{Sum of Valid Temperature Samples}}{\text{Number of Valid Samples}} \qquad \text{…….. (15)}$$

After averaging, an exponential moving average filter was applied to smooth the temperature value. Used alpha as 0.5 and the filtered DS18B20 value was used as the final growing-medium temperature. This value was used for the root zone cooling decision and was included in the IR feedback packet.

2.8.4 BH1750FVI Light Sensor Calibration and Filtering

The BH1750FVI light sensor [35] was used to measure the light intensity of the grow light. Before using the sensor value in the control node, calibration was carried out using a reference lux meter. This was done because the BH1750FVI sensor reading was different from the lux meter reading. First, the BH1750FVI raw digital value was converted into lux [35]. Lux calculation:

$$\text{BH1750 Lux} = \frac{1.2}{\text{Raw Value}} \qquad \text{…….. (16)}$$

For each calibration point, five BH1750FVI readings were taken. These five readings were averaged to reduce small reading variations:

$$\text{Average BH1750 Lux} = \frac{L1+L2+L3+L4+L5}{5} \qquad \text{…….. (17)}$$

The average BH1750FVI lux value was compared with the reference lux meter value.

The calibration factor:

$$\text{Calibration Factor} = \frac{\text{Average BH1750 Lux}}{\text{Lux Meter Reading}} \qquad \text{…….. (18)}$$

The BH1750FVI sensor readings were lower than the reference lux meter readings. Therefore, a correction factor was applied to the BH1750FVI reading. The average calibration factor obtained from the experiment was approximately 1.75.

The final calibrated lux value was calculate using:

$$\text{Calibrated Lux} = \frac{\text{Raw Value}}{1.2} \times 1.75 \qquad \text{…….. (19)}$$

In MSP430 code, integer arithmetic was used instead of floating-point calculation. Therefore, the lux value was stored in (x10) format, where the actual lux value is multiplied by 10. The conversion and calibration were implemented.

After calibration, filtering was applied to make the light intensity value more stable. The BH1750FVI reading can slightly change due to light flickering, sensor noise, and small

changes in sensor angle. Therefore, five consecutive readings were first averaged. After that, an Exponential Moving Average filter was applied. The final light measurement process was raw BH1750FVI reading, lux conversion, averaging, calibration and EMA filtering [37]. This process provided a more accurate and stable light intensity value for the control node.

2.9 Control Node Sensor Processing Sequence

The sensor reading sequence was initiated by a VLC command from the grow light unit. The sensors were read in a fixed order to make the data acquisition process repeatable. First, the capacitive soil moisture sensor was read through the MSP430 ADC input. This sensor produced an analogue voltage according to the moisture condition of the growing medium, and the ADC converted this voltage into a digital value for processing. Next, the BH1750FVI light sensor was read through the I2C interface to obtain the light intensity reaching the plant canopy.

After the light measurement, the SHT3X sensor was read through the I2C interface to obtain air temperature and relative humidity around the plant canopy. Finally, the DS18B20 sensor was read to obtain the growing medium temperature near the root zone. This value was used to monitor the root zone condition and to support the cooling control decision.

After all sensor readings were collected, the MSP430 processed the measured values and compared them with the selected strawberry control thresholds. Based on these comparisons, the firmware updated the actuator control states. The sensor readings and actuator states were then packed into the IR feedback frame and transmitted back to the grow light unit through the IR uplink.

Different sample numbers were used for different sensors because each sensor had different signal behavior and conversion characteristics. After completing the sensor reading, actuator update and IR feedback transmission, the control node returned to low power mode and waited for the next VLC wake-up and command cycle.

The return to sleep sequence

1. Complete actuator decision and IR feedback transmission.

2. Turn OFF switched sensors.
3. Release the I2C bus.
4. Turn OFF actuator outputs after timed operation.
5. Disable the ADC module.
6. Enter LPM3 sleep mode.
7. Wake after the selected sleep interval.
8. Re-enable the ADC.
9. Start the next sensing cycle.

This sequence reduced unnecessary active mode energy consumption and allowed the passive powered control node to operate only when sensing or control action was required.

2.10 Environmental Control Actuators

The proposed VLC passive powered sensing and control module utilizes multiple actuators to regulate environmental conditions within the cultivation environment. These actuators operate in response to sensor measurements and control decisions generated by the MSP430 microcontroller, enabling automated regulation of relative humidity, air temperature, and soil moisture. The actuator subsystem consists of a 5 V ultrasonic humidifier, a 12 V DC cooling fan, and a 12 V solenoid valve.

The ultrasonic humidifier was incorporated to regulate relative humidity within the cultivation environment. Automated humidity regulation is commonly employed in controlled-environment agriculture to maintain suitable growing conditions and reduce environmental stress on plants [15].

A DC cooling fan was integrated into the system to regulate air temperature and improve air circulation. The fan is activated when elevated temperature conditions are detected, allowing excess heat to be removed from the cultivation environment. Ventilation systems are widely used in greenhouse and indoor farming applications to maintain stable microclimatic conditions and improve environmental control performance [38]. The

selected AD0812MS-C70 fan operates at a rated voltage of 12 V DC with a rated current of 0.16 A and provides an airflow of approximately 21.8 CFM [39].

Soil moisture regulation was achieved by use of a normally closed 12 V DC solenoid valve. The valve controls the flow of water supplied to the irrigation system and is activated whenever soil moisture measurements indicate insufficient water availability within the growing medium. Automated irrigation systems are widely adopted in smart agriculture applications because they improve water use efficiency while ensuring adequate water availability for plant growth. The selected solenoid valve operates at 12 V DC with a current consumption of approximately 320 mA [40].

The integration of these actuators enables the system to automatically regulate environmental conditions based on real-time sensor measurements. By controlling humidity, temperature, and irrigation processes, the actuator subsystem contributes to the maintenance of favorable growing conditions while reducing manual intervention and improving resource utilization efficiency.

2.10. 1 Soil Temperature control using Thermoelectric cooling

The TEC1-12705 thermoelectric cooling module can draw up to 5.3 A at a maximum operating voltage of 14.2 V [41]. Operating the Peltier module at its maximum current was not necessary for the proposed soil cooling application and could result in excessive localized cooling near the root zone. Such conditions may produce non uniform soil temperature distributions and potentially expose plant roots to undesirable low temperature conditions. Therefore, a current control mechanism was implemented to regulate the cooling intensity of the thermoelectric module.

A DC-DC buck converter was modified and configured for constant current operation by replacing the current sensing resistor with a 1.2 Ω, 2 W resistor. The current limiting operation of the converter is based on the voltage developed across the sensing resistor.

The current output can be approximate using:

$$I = \frac{V_{ref}}{R_{sense}} \qquad \text{…….. (20)}$$

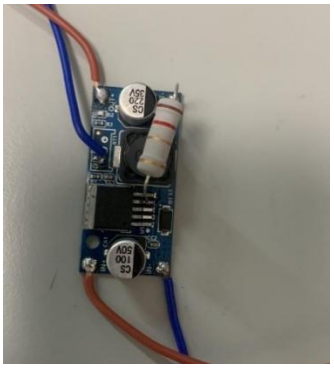


Figure 2.11 : Modified DC–DC buck converter configured for constant-current operation using a 1.2 Ω, 2 W sensing resistor

Therefore, the modified converter limited the operating current of the TEC1-12705 module to approximately 1.03 A. This current level was selected to provide sufficient cooling performance while preventing excessive localized cooling of the surrounding soil.

2.10. 2 Actuator Driver Circuit Design

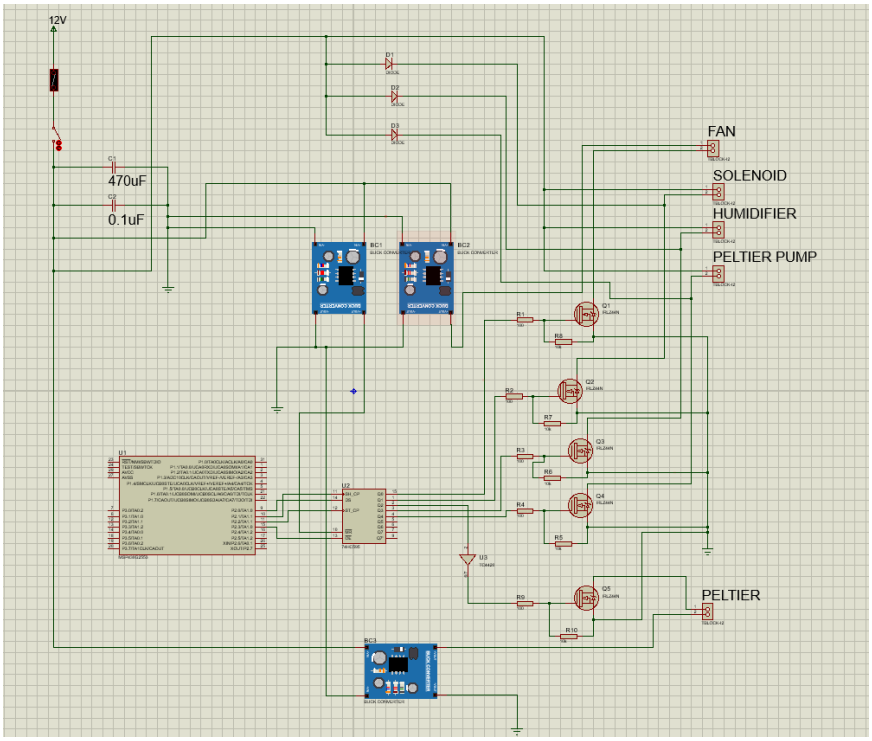


Figure 2.12: Schematic diagram of the actuator driver circuit showing the MSP430G2553 microcontroller, SN74HC595 shift register, IRLZ44N MOSFET switching stage, TC4420 gate driver, buck converters, and environmental actuators

The actuator driver circuit was designed to connect the MSP430G2553 microcontroller to the actuators used in the environmental control system. The MSP430 output pins cannot directly drive loads such as the fan, humidifier, pump, solenoid valve, and Peltier module

because these actuators require high power than the microcontroller can supply. Therefore, a separate driver circuit was used to switch the actuators safely. The circuit uses an SN74HC595 shift register to expand the number of control outputs from the MSP430, IRLZ44N MOSFETs to switch the actuator loads, a TC4420 MOSFET gate driver for the Peltier cooling channel, and DC-DC buck converters to provide the required supply voltages.

The TEC1-12705 thermoelectric cooling module represents the highest current load within the actuator subsystem and therefore required a dedicated driver stage. A TC4420 MOSFET gate driver was incorporated between the SN74HC595 output and the IRLZ44N power MOSFET controlling the Peltier channel.

The TC4420 provides high current gate drive capability, enabling rapid charging and discharging of the MOSFET gate capacitance [42]. This improves switching performance, reduces switching losses, and ensures reliable operation of the thermoelectric cooling channel. The output of the TC4420 driver was connected to the gate of the IRLZ44N MOSFET, which controlled power delivery to the TEC1-12705 module through the current-controlled buck converter described in Section 2.11.1.

To support the control of multiple actuators while minimizing pin utilization, an SN74HC595 serial-in parallel-out shift register was incorporated into the design [43]. The shift register converts serial data transmitted from the microcontroller into parallel output signals, allowing multiple actuators to be controlled using only three microcontroller pins.

The outputs of the SN74HC595 were assigned to individual actuator channels, enabling independent control of the humidifier, cooling fan, solenoid valve, water pump, and thermoelectric cooling system. This approach simplified the hardware design and provided flexibility for future expansion of the actuator subsystem.

IRLZ44N N-channel MOSFETs were used as low side switching devices for actuator control [44]. MOSFETs allow high-current loads to be controlled using low power digital signals generated by the MSP430 microcontroller through the SN74HC595 shift register.

To protect the switching devices from voltage transients generated by inductive loads, flyback diodes were connected across the solenoid valve and water pump. These protection

components suppress reverse voltage spikes produced during switching operations and improve the reliability of the actuator driver circuit.

2.11 IR Uplink implementation

2. 11. 1 IR Feedback Communication Method

The control node used infrared communication as the feedback path from the node to the grow light unit. The IR feedback method was selected because the control node only needed to send a small amount of information after each sensing cycle. It did not require continuous high-speed communication. Short range infrared communication is suitable for indoor and enclosed applications where low-cost optical communication is required and infrared links are commonly implemented using intensity modulation and direct detection.

In the proposed method, the IR transmitter was placed on the control node. The MSP430 generated the IR data packet and transmitted it through the KY-005 infrared transmitter module [45]. The transmitter did not send a steady infrared light signal. Instead, it used a 38 kHz carrier during active pulse periods and switched the carrier off during space periods. This made the communication signal easier for a demodulating IR receiver to detect and helped separate it from normal background light [46].

The feedback packet was designed as a fixed length 80-bit frame. This packet included the grow light ID, node ID, five sensor values, four actuator status bits and one CRC error-detection byte. The fixed packet length made the transmitter code simpler because the MSP430 always sent the same frame format after every sensing cycle. To improve reliability, the same packet was transmitted three times with a short delay between transmissions. The IR uplink was used only after the control node completed one sensing and control cycle.

Indoor IR links can still be affected by distance, alignment, shadowing and ambient light noise so these conditions were considered during the IR reliability testing [47].

The KY005 module was connected to the MSP430. That GPIO pin was switched on and off rapidly to generate the IR carrier signal during each mark period. Therefore, the data was not sent as continuous infrared light. It was sent as timed IR bursts.

The IR LED used in this type of transmitter operates in the near infrared range. A 940 nm IR LED is suitable for short range free air transmission because it is commonly used in infrared remote control and applied IR systems. The IR LED direction was important because the received optical power reduces when the receiver is not aligned with the transmitter. Therefore, during testing, the transmitter was pointed towards the grow light receiver [48].

The receiver side used a KY022 type demodulating IR receiver module. This receiver was placed near the grow light unit. It detected the incoming 38 kHz IR bursts and produced a digital output signal for the receiver controller. The receiver did not need to measure the raw optical carrier directly because the receiver module already included internal filtering and demodulation. The VS1838B receiver datasheet states that the module contains a photodetector and preamplifier, has an internal filter, improves immunity against ambient light and gives a demodulated output that can be decoded by a microprocessor [45].



Figure 2.13 - IR Waveforms for transmitter output and the receiver output

2. 11. 2 IR Uplink Frame Structure

The IR feedback frame was designed as a fixed length 80bit packet. A standard short IR command frame was not enough for this project because the control node had to transmit several sensor readings and actuator states. Therefore, a custom 80bit packet was used with NEC style timing.

The packet contained four main parts such as identification bits, sensor data, actuator status bits and an error checking byte. The identification bits were used to identify the target grow light unit and the transmitting control node. The sensor data fields were used to send the final processed sensor values. The actuator bits were used to show the actuator decisions made by the MSP430. The CRC byte was used to detect packet errors.

Table 2-7 : Field definitions of the 80-bit IR packet

Field	Size	Description
Grow-light ID	4 bits	Identifies the target grow-light unit
Node ID	4 bits	Identifies the transmitting control node
Sensor 1	12 bits	Soil moisture value
Sensor 2	12 bits	Air temperature value
Sensor 3	12 bits	Relative humidity value
Sensor 4	12 bits	Light intensity value
Sensor 5	12 bits	Growing-medium temperature value
Actuator 1	1 bit	Fan status
Actuator 2	1 bit	Irrigation status
Actuator 3	1 bit	Humidifier status
Actuator 4	1 bit	Peltier cooling system status
CRC-8	8 bits	Error-detection value

Each sensor value was allocated 12 bits. A 12-bit field can represent values from 0 to 4095. This was enough for the scaled sensor values used in the control node. The actuator status values were stored as single bit values. A value of 1 meant that the actuator was requested on, and a value of 0 meant that the actuator was off. The 80bit packet was stored as 10 bytes before transmission. The first 9 bytes contained the ID, sensor and actuator information. The final byte contained the CRC value. This packet structure kept the

feedback frame compact while still carrying all required information from one sensing cycle.

2. 11. 3 NEC Style Timing Method

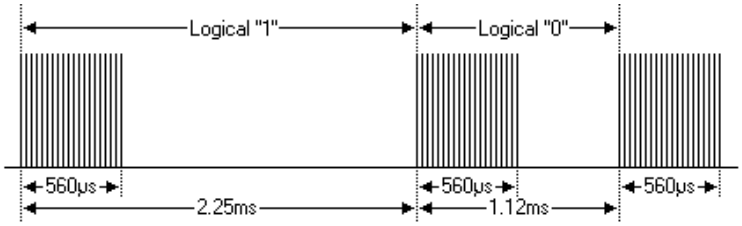


Figure 2.14 : NEC timing diagram

The IR packet was transmitted using NEC style pulse distance timing. This method was selected because it is simple to generate using the MSP430 and can be detected by common demodulating IR receiver modules. In this project, the standard NEC idea was not used as a normal remote-control command. Instead, the same timing style was used to transmit the custom 80bit feedback packet.

The transmitter used a 38 kHz carrier during each mark period. During a mark, the MSP430 rapidly switched the IR transmitter pin on and off to create the carrier. During a space, the IR transmitter was switched off. Therefore, the transmitted signal was made from carrier bursts and silent spaces.

The packet started with a leader mark and leader space. The leader mark was 9000 µs, and the leader space was 4500 µs. This start pattern helped the receiver identify the beginning of a new packet. After the leader section, the 80 data bits were transmitted one by one. Each bit started with a 560 µs mark. The value of the bit was decided by the length of the space after the mark. A logical 0 used a 560 µs space, while a logical 1 used a 1690 µs space. These values follow the NEC pulse-distance timing concept, where a 38 kHz pulse carrier and different space durations are used to represent logical 0 and logical 1 [46].

Table 2-8 : NEC style timing values used for IR transmission

Signal part	**Timing value**
Carrier frequency	Approximately 38 kHz
Leader mark	9000 µs

Signal part	Timing value
Leader space	4500 µs
Bit mark	560 µs
Logical 0 space	560 µs
Logical 1 space	1690 µs
Final mark	560 µs
Inter-burst gap	40 ms

The MSP430 generated the carrier in software. At the 1 MHz clock setting, the IR pin was kept high for about 13 µs and low for about 13 µs. This produced a carrier period of about 26 µs, which is close to 38 kHz. The 80bit packet was transmitted in MSB first order. The first transmitted bit was the most significant bit of the grow light ID. This timing method allowed the control node to transmit a structured IR packet using only one GPIO pin and software timing.

2. 11. 4 CRC Error Detection

A CRC error-detection byte was added to the end of the IR packet to check whether the received packet was correct. Distance, angle, obstruction and surrounding light effects can cause some bits to be missed or decoded wrongly. Therefore, the packet required an error checking method before the received data was accepted. The program used the CRC-8/MAXIM method. An 8bit CRC was selected because the packet was short and the MSP430 had limited memory and processing resources. CRC checking is stronger than a simple parity bit because it can detect more types of bit errors in a data frame. At the receiver side CRC was calculated again using the received first 9 bytes. The recalculated CRC was then compared with the CRC byte received in Byte 9. If both values matched, the packet was considered valid. If the values did not match, the packet was rejected. This method helped prevent corrupted sensor readings or actuator status bits from being accepted by the grow light unit. Therefore, the CRC field improved the reliability of the IR feedback communication [49].

2.12 Enclosure design

A custom enclosure was designed to house the sensors, MSP430-based control circuitry, and associated electronic components of the control node. The enclosure was designed with dimensions of 12 cm × 11 cm × 6 cm to provide sufficient space for the electronic components while maintaining a compact design. Internal clip mechanisms were incorporated to securely hold the PCB. Sensor mounting locations were integrated into the enclosure lid to facilitate environmental monitoring. Additionally, protective housing was designed for the SHT3X sensor to prevent physical damage while maintaining exposure to ambient conditions. The enclosure was fabricated using a 3D printing process.

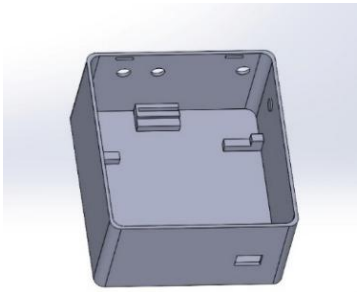


Figure 2.15 : CAD design of Box protects PCB circuitry and sensors

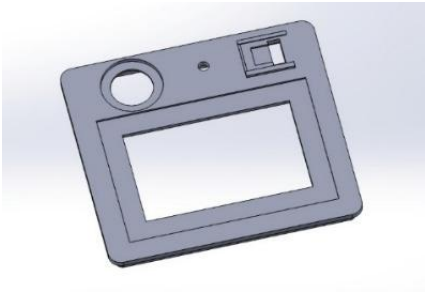


Figure 2.16 : CAD design lid for the sensor node to hold sensors and cover the circuitry parts

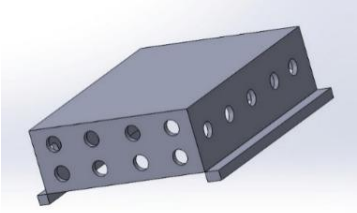


Figure 2.17 : CAD design for the temperature sensor protection

CHAPTER 3. Results and Discussion

3.1 VLC Performance analysis

3.1.1 Frequency Response and FSK Frequency Selection Analysis

The frequency response of the solar panel-based VLC receiver path was evaluated to identify suitable frequencies for Binary FSK command transmission. Figure 3.1 shows the measured peak to peak voltage variation with modulation frequency at different transmitter - receiver distances.

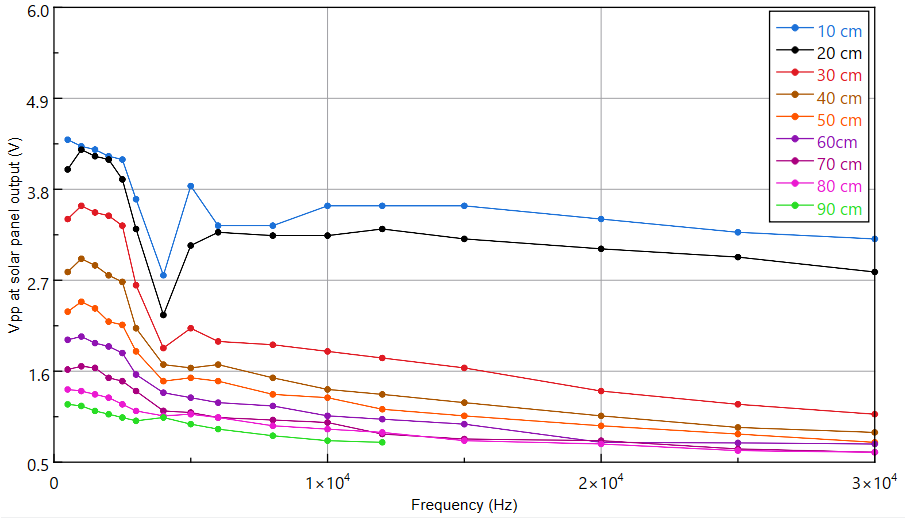


Figure 3.1 : Peak to peak voltage at solar panel output (V) vs. Frequency

The results show that the received AC signal level decreases as the transmitter - receiver distance increases. This is because the optical power incident on the solar panel decreases with distance. The results also show that the low kHz frequency region produces a stronger received signal than the higher frequency region. This is mainly due to the limited frequency response of the solar-panel receiver and the loading effect of the receiver front-end.

During the initial VLC tests, lower Binary FSK frequencies of 800 Hz for logic 0 and 1000 Hz for logic 1 were used. These frequencies were detectable by the solar panel receiver, but the separation between the two frequencies was relatively small, which reduced the pulse counting margin at the MSP430 receiver. The frequency pair was then improved to 1.5 kHz for logic 0 and 2 kHz for logic 1, while 4 kHz was used as idle frequency. This configuration produced more stable decoding at shorter distances because the received

signal amplitude was still strong in the low kHz region and the frequency separation was more suitable for timer-based pulse counting. However, this low frequency configuration limited the achievable data rate and was not suitable for VLC communication at longer distances.

Therefore, a high frequency Binary FSK configuration was tested for improved data rate performance further. For the final high data rate VLC test, the frequency pair was changed to 3 kHz for logic 0 and 5 kHz for logic 1, while 10 kHz was used as the idle frequency. This configuration supported a shorter 5 ms symbol duration, corresponding to a data rate of 200 bps. The larger separation between the data frequencies improved symbol classification during pulse count-based decoding and enabled longer distance communication compared with the earlier low frequency configurations.

However, at very short distances below approximately 30 cm, decoding was not reliable due to excessive received signal level and receiver saturation effects. This indicates that the optimum frequency set depends not only on the solar-panel frequency response but also on the received signal amplitude and transmitter - receiver distance.

A local reduction in received Vpp was observed around 4 kHz in the measured frequency-response curve. Since the signal level increased again at nearby higher frequencies, this reduction cannot be explained only as the normal low pass behavior of the solar panel. It is caused by the combined response of the both transmitter driver and solar panel. Therefore, the frequency response result was used as a practical guide for selecting stable operating frequencies rather than as an isolated solar panel bandwidth measurement.

Based on the frequency response test and practical decoding experiments, the final VLC downlink used 3 kHz for logic 0, 5 kHz for logic 1, and 10 kHz as the idle state. This configuration provided a practical balance between received signal detectability, frequency separation, and data rate improvement.

3.1.2 VLC Packet Error Rate and Data Rate Selection Analysis

The packet error rate test was conducted to evaluate the reliability of the VLC downlink used in the passive powered control node. During the test, the transmitter repeatedly sent VLC test frames using Binary FSK modulation. After frequency selection experiments, the

final FSK frequency set used 3 kHz for binary 0, 5 kHz for binary 1, and 10 kHz for the idle state. The VLC link was then tested at three data rates: 50 bps, 100 bps, and 200 bps. These data rates correspond to symbol durations of 20 ms, 10 ms, and 5 ms, respectively.

For each transmitter - receiver distance, 1000 VLC frames were transmitted. Each frame contained start symbols, a synchronization byte, address byte, control byte, data byte, and CRC byte. The receiver decoded the incoming frame using the MSP430 Timer_A pulse-counting method. A packet was accepted only when synchronization was detected and the CRC validation was successful.

PER is calculated in different distances using:

$$\text{PER} = \frac{\text{No of failed packets}}{\text{Total no of transmitted packets}} \times 100\% \qquad \ldots\ldots\ldots\ldots (21)$$

The variation of packet error rate (PER) with transmitter - receiver distance for the 50 bps, 100 bps, and 200 bps VLC configurations is shown in Figure 3.2.

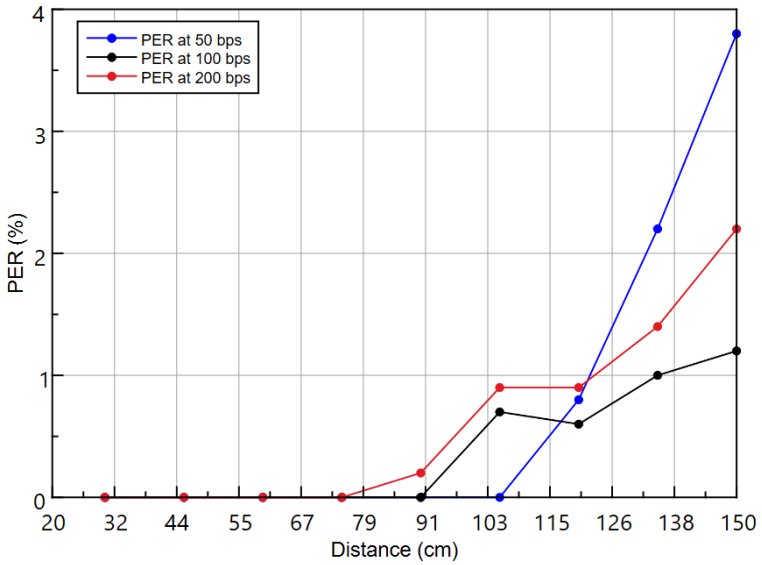


Figure 3.2 : PER (%) vs. Distance

The 50-bps configuration provided the longest symbol duration (20ms) and the largest timing margin for pulse counting. It achieved 0% PER up to 105 cm. However, the packet error rate increased at longer distances, reaching 0.8% at 120 cm, 2.2% at 135 cm, 3.8% at 150 cm and 6.5% at 165 cm. Therefore, increasing the symbol duration alone does not guarantee the best long-distance performance, because the receiver performance also

depends on synchronization, signal amplitude, and the stability of the detected optical waveform.

The 100-bps configuration provided strong reliability over the tested distance range. It achieved 0% PER up to 90 cm, and the PER remained low at longer distances, with 0.7% at 105 cm, 0.6% at 120 cm, 1.0% at 135 cm and 1.2% at 150 cm. This result shows that the 10 ms symbol duration provided sufficient pulse counting margin while reducing the frame duration compared with the 50-bps configuration.

The 200-bps configuration used the shortest symbol duration of 5 ms and therefore provided the fastest frame transmission among the tested data rate configurations. It achieved 0% PER up to 75 cm. At longer distances, the PER increased gradually, with 0.2% at 90 cm, 0.9% at 105 cm, 0.9% at 120 cm, 1.4% at 135 cm and 2.2% at 150 cm. Although the error rate was higher than the 100-bps configuration at longer distances, the 200-bps configuration still supported successful command-frame reception up to 150 cm with a relatively low packet error rate.

The use of Binary FSK helped improve the reliability of the receiver because the MSP430 decoded the received signal based on frequency pulse count rather than only signal amplitude. This is useful in a solar panel receiver, where the received signal amplitude can change due to light intensity, panel alignment, distance, and loading from the energy harvesting path.

The CRC validation prevented corrupted frames from being accepted as valid control commands. Therefore, even when packet errors occurred, invalid frames were rejected by the receiver instead of being used.

3.2 Solar panel and energy harvesting system performance analysis

3.2.1 Solar Panel V-I Characteristics

Since the solar panel used in the control node did not include a complete manufacturer datasheet, its electrical behavior was experimentally evaluated using a variable resistive load. Figure 3.3 shows the measured V-I characteristics of the solar panel under two illumination conditions. The curves show the relationship between the terminal voltage and output current of the panel.

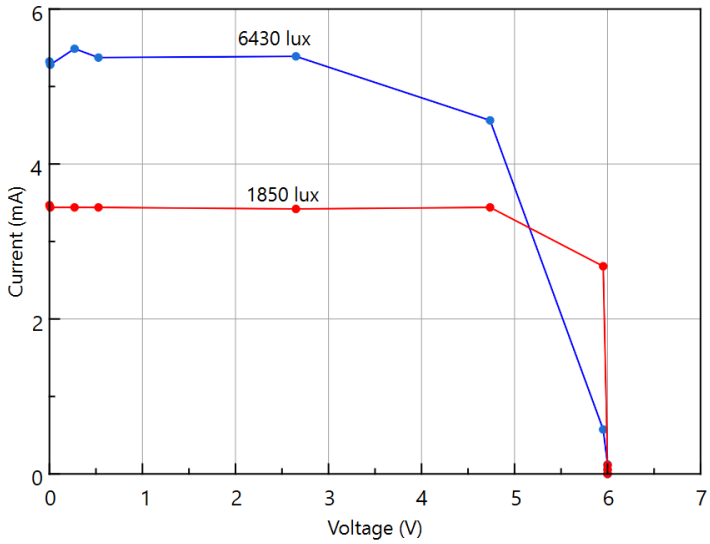


Figure 3.3 : V-I curve of the solar panel under different intensity levels

At the short-circuit condition, the terminal voltage is approximately zero and the current reaches its maximum value. At the open-circuit condition, the output current is approximately zero and the terminal voltage reaches its maximum value. For the 6430-lux condition (100% duty cycle at 60 cm distance), the measured short-circuit current was approximately 5.32 mA and the open-circuit voltage was approximately 6.0 V. For the 1850 lux condition (100% duty cycle at 80 cm distance),, the measured short-circuit current was approximately 3.47 mA and the open-circuit voltage was approximately 5.76 V. This shows that the short-circuit current decreased significantly when the received illumination level was reduced, while the open-circuit voltage changed only slightly.

The output power of the solar panel was calculated from the measured voltage and current values. Figure 3.4 and Figure 3.5 shows the corresponding P-V characteristics of the solar

panel. The peak point of each P-V curve represents the maximum power point (MPP) of the solar panel under the tested illumination condition.

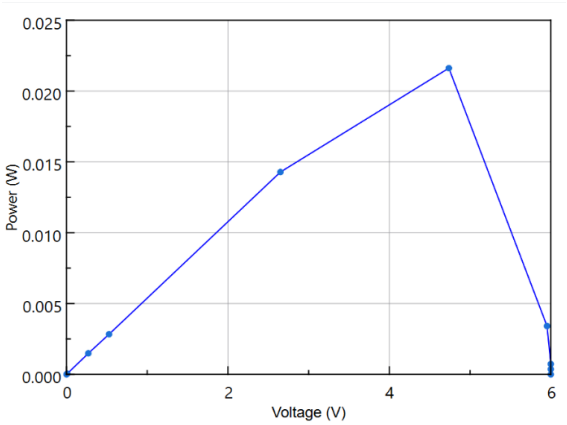


Figure 3.5 : P-V curve of the solar panel under 1850 Lux

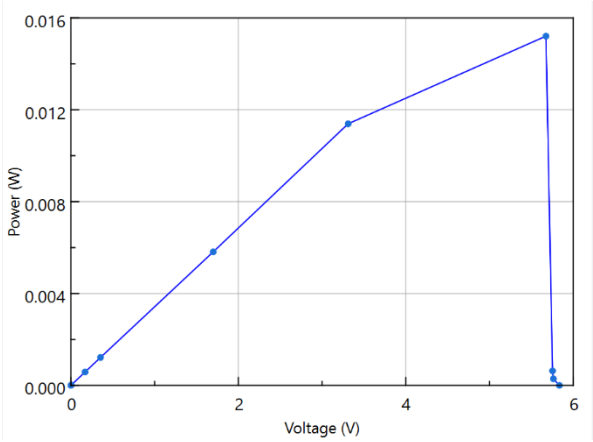


Figure 3.4 : P-V curve of the solar panel under 6430 Lux

For the 6430-lux condition, the maximum output power was approximately 21.6 mW at about 4.74 V and 4.56 mA. For the 1850 lux condition the maximum output power was approximately 15.2 mW at about 5.67 V and 2.68 mA. Therefore, when the illumination level decreased from 6430 lux to 1850 lux, the available maximum power decreased by approximately 29.6%. The harvested power available to the energy harvesting module strongly depends on the received optical intensity.

The measured V-I and P-V characteristics were used to understand the practical behavior of the selected solar panel and to support the design of the harvesting path. Since the control node operates from harvested optical energy stored in a supercapacitor, this characterization was necessary to estimate whether the panel could provide sufficient

energy for the low-power electronics, including the MSP430 microcontroller, VLC receiver front-end, sensors, and IR uplink circuit.

3.2.2 Supercapacitor Charging and Practical Operating Cycle Analysis

The supercapacitor charging behavior was analyzed to evaluate the energy harvesting performance of the proposed passive-powered control node. Two charging conditions were considered in this analysis. First, the storage-side charging time was measured from approximately 2.93 V to 4.245 V, where 4.245 V was the maximum storage voltage reached by the BQ25570 harvesting module. Second, the practical cyclic operating behavior was evaluated between approximately 2.93 V and 3.33 V. In the practical cycle, the BQ25570 module turned off the regulated output when the storage voltage decreased to approximately 2.93 V and enabled the regulated 3.3 V output when the storage voltage reached approximately 3.33 V.

3.2.2.1 Storage-Side Charging Time

The storage-side charging test was carried out to measure the time required for the supercapacitor to charge from approximately 2.93 V to 4.245 V under different transmitter–receiver distances and LED duty cycles. This test represents the full charging capability of the energy-harvesting path when the storage capacitor is allowed to charge up to the maximum voltage reached by the module. Charging times greater than one hour were not considered because they are not practical for the intended low-power control-node operation. Figure 3.6 shows the variation of charging time with distance for different LED duty cycles.

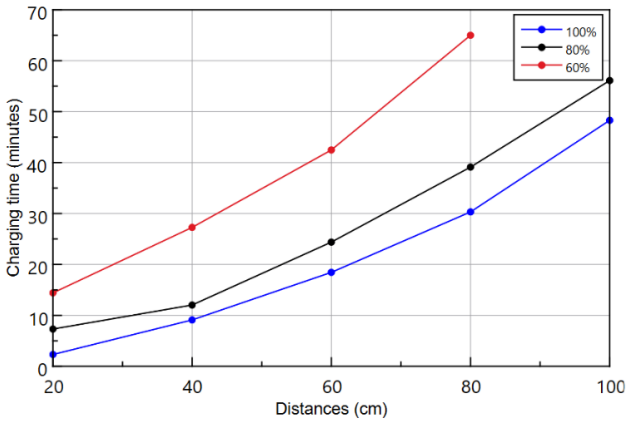


Figure 3.6 : Supercapacitor charging time from 2.93 V to 4.245V at different distances and LED duty cycles

The results show that the charging time increased as the distance between the light source and the solar panel increased. This occurred because the received optical intensity decreased with distance, reducing the harvested current available to the energy-harvesting module. The charging time also increased when the LED duty cycle was reduced, because lower duty cycles provide lower average optical power to the solar panel.

At 100% duty cycle, the charging time increased from approximately 2.5 min at 20 cm to approximately 48.5 min at 100 cm. At 80% duty cycle, the charging time increased from approximately 7.5 min at 20 cm to approximately 56.2 min at 100 cm. At 60% duty cycle, the charging time was approximately 14.7 min at 20 cm, 27.4 min at 40 cm, and 42.8 min at 60 cm. These results confirm that the full storage charging time depends strongly on received illumination level, transmitter–receiver distance, and LED duty cycle.

3.2.2.2 Practical Charge - Discharge Cycle and Next Operation Time

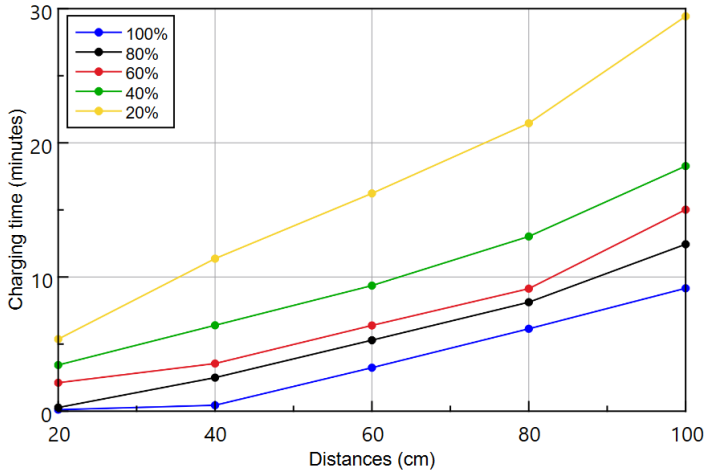


Figure 3.7 : Practical supercapacitor charging time between 2.93 V and 3.33 V

Although the supercapacitor could charge up to approximately 4.245 V during the storage - side charging test, the practical operation of the control node occurred over a smaller voltage window. During loaded operation, the BQ25570 module enabled the regulated 3.3 V output when the storage voltage reached approximately 3.33 V. At this point, the MSP430 control circuit and VLC receiver front-end started operating and consuming energy from the supercapacitor. If the harvested input power was lower than the active load power, the supercapacitor voltage decreased. When the storage voltage dropped to approximately 2.93 V, the regulated output turned off and the capacitor began charging again. Therefore, the practical next operation time is determined by the recharge time from approximately 2.93

V to 3.33 V, not by the full charging time from 2.93 V to 4.245 V. The measured practical-cycle results show that the recharge time increased with distance and decreased with higher LED duty cycle. At 100% duty cycle, the recharge time increased from approximately 10.7 s at 20 cm to approximately 9.25 min at 100 cm. At 80% duty cycle, the recharge time increased from approximately 27.0 s at 20 cm to approximately 12.7 min at 100 cm. At 60% duty cycle, the recharge time increased from approximately 2.2 min at 20 cm to approximately 15.0 min at 100 cm.

These results show that the proposed control node operates as a cyclic energy-harvesting system under many practical conditions. Therefore, the next operation time depends on the recharge duration between the output-disable and output-enable voltage levels. This behavior confirms the importance of reducing active-mode current consumption.

3.3 Sensor Output Results

3.3.1 Capacitive Soil Moisture Sensor calibration results

The capacitive soil moisture sensor was tested by collecting 100 samples from the MSP430 ADC input. The sensor produced an analogue output so small variations were expected in the raw ADC readings.

The raw ADC values were mostly between 682 and 687. However, a few sudden changes were observed. These sudden changes affected the raw moisture percentage because the moisture calculation was based on the ADC value. The dry reference ADC value was 820, and the wet reference ADC value was 560. Using these calibration values the raw ADC readings were converted into relative moisture percentage. The raw moisture values were mostly around 51% to 53%. However, due to sudden raw ADC changes, the raw moisture value dropped to 47.6% at sample 7 and increased to 58.4% at sample 39.

This result shows that the sensor gave a generally stable moisture reading, but the analogue signal still had some sudden noise. Therefore, filtering was needed before using the moisture value for irrigation control.

Table 3-1 : Summary of the soil moisture values

Parameter	Minimum	Maximum	Average

Raw ADC	668	696	684.68
Raw moisture percentage	47.60%	58.40%	52.00%
EMA moisture percentage	51.60%	53.00%	52.20%

The result shows that the raw ADC signal had a wider range because of a few abnormal readings. After filtering, the moisture value stayed in a smaller and more stable range. This made the sensor output more suitable for actuator decision-making.

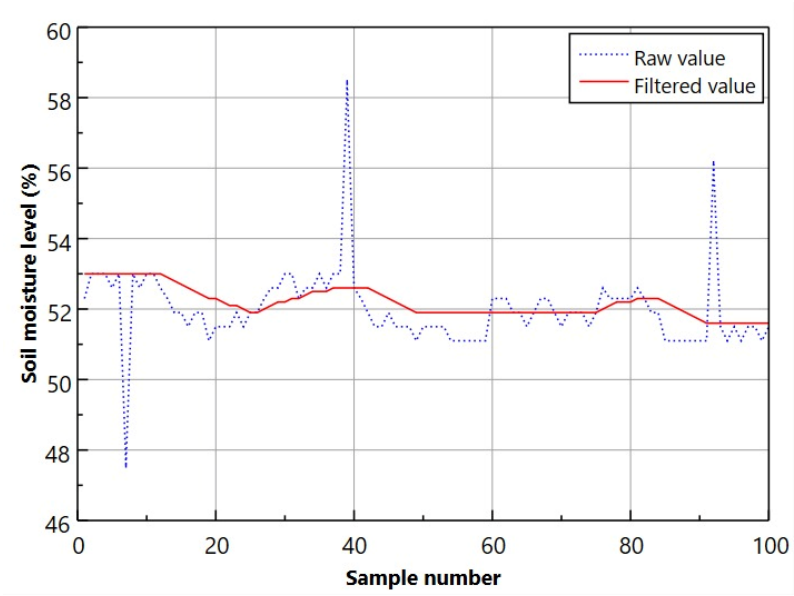


Figure 3.8 : Soil moisture level vs sample number

The soil moisture calibration converted the raw ADC value into relative moisture percentage. The dry reference value was measured in air and was taken as 0% relative moisture. The wet reference value was measured in water and was taken as 100% relative moisture. Therefore, the calibration equation was:

$$Relative\ moisture\ (\%) = \frac{820 - Measured\ ADC}{260} \times 100 \qquad \text{..... (22)}$$

The test result showed that the moisture value was mainly around 52%. This means the measured condition was between the dry and wet reference points. Since the dry reference was measured in air and the wet reference was measured in water, this value was considered a relative moisture index, not an absolute volumetric water content value.

Filtering clearly improved the soil moisture result. The raw moisture value changed from 47.6% to 58.4%, which gave a total range of 10.8%. After the ADC filtering stage, the

filtered moisture value stayed between 51.1% and 53.4%, giving a smaller range of 2.3%. After EMA filtering, the final moisture value stayed between 51.6% and 53.0%, giving a range of only 1.4%. This shows that the filtering method reduced sudden noise in the analogue sensor signal. The abnormal readings at samples 7, 39, and 92 did not strongly affect the final EMA moisture value. The EMA output changed slowly and smoothly from about 53.0% at the beginning to about 51.6% at the end of the test. The raw moisture line has sudden changes at a few sample points, while the filtered and EMA lines remain smoother. This confirms that the filtering process made the soil moisture reading more stable.

3.3.2 SHT3X Temperature and Humidity Sensor calibration results

3.3.2.1 Temperature Calibration and Filtering Result

The SHT3X sensor was tested by collecting 100 air temperature samples. The sensor provided raw temperature count values through the I2C interface, and these values were converted into temperature in degrees Celsius. During the test, the raw temperature count changed from 26099 to 26125. After conversion, the air temperature stayed between 24.69 °C and 24.76 °C. The average air temperature was approximately 24.72 °C. This shows that the air temperature around the sensor was stable during the test period.

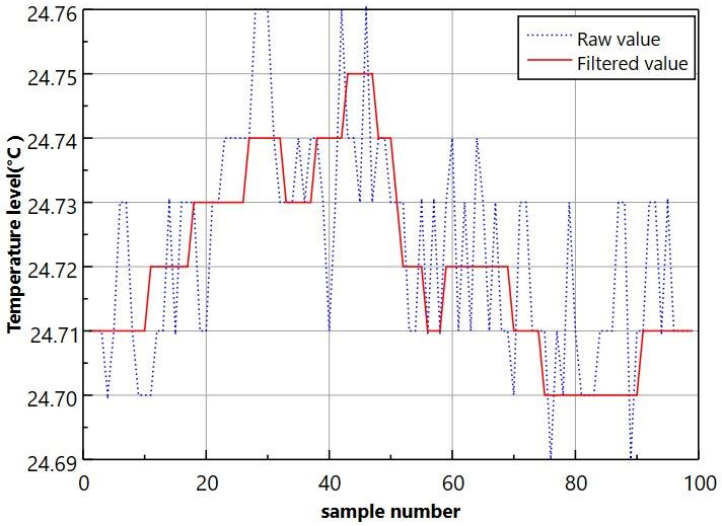


Figure 3.9 : Temperature vs. sample number

Table 3-2 : summery of Temperature Readings

Parameter	Minimum	Maximum	Average
Raw temperature count	26099	26125	26112.78
Raw air temperature	24.69 °C	24.76 °C	24.72 °C
EMA filtered air temperature	24.70 °C	24.75 °C	24.72 °C

The raw air temperature varied between 24.69 °C and 24.76 °C. Although this variation was small, filtering was still useful because actuator control should not depend on sudden small changes in one reading. The EMA filtered temperature was smoother than the raw temperature. The filtered value changed between 24.70 °C and 24.75 °C. The raw temperature range was 0.07 °C, while the EMA filtered range was reduced to 0.05 °C.

Raw temperature line changes more frequently, while the EMA line changes more slowly. The filtered line follows the general trend of the temperature but it does not react strongly to every small raw reading change. The result shows that the SHT3X air temperature reading was already stable, but EMA filtering made the final value smoother for control use. The filtered temperature was suitable for actuator decision-making and IR feedback transmission.

3.3.2.1 Humidity Calibration and Filtering Result

The SHT3X sensor was tested by collecting 100 relative humidity samples. The sensor produced raw humidity count values through the I2C interface, and these values were converted into relative humidity percentage.

During the test, the raw humidity count changed from 39601 to 40040. After conversion, the relative humidity changed from 60.42%RH to 61.09%RH. The average humidity value was approximately 60.78%RH. This shows that the humidity around the sensor was fairly stable during the test.

Table 3-3 : Summery of humidity readings

Parameter	Minimum	Maximum	Average
Raw humidity count	39601	40040	39837.12
Raw relative humidity	60.42%RH	61.09%RH	60.78%RH
EMA filtered relative humidity	60.48%RH	61.03%RH	60.78%RH

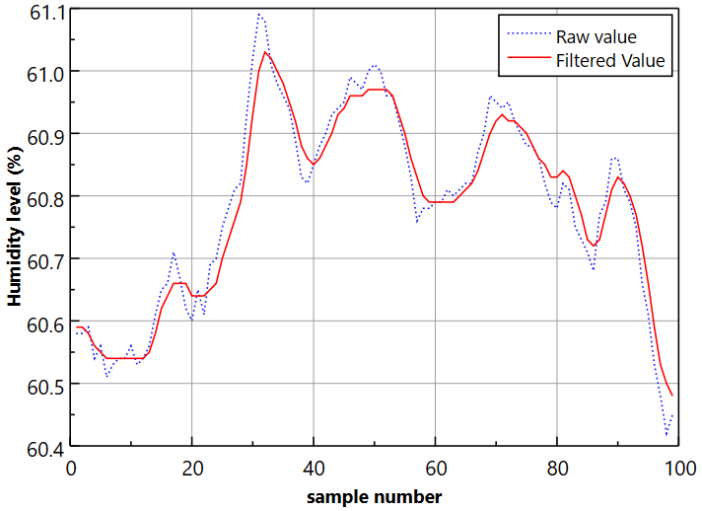


Figure 3.10 : Humidity vs. sample number

The SHT3X humidity readings were filtered using an exponential moving average filter. The graph shows that the raw humidity line changed more quickly, while the EMA line changed more smoothly. The raw humidity value changed from 60.42% RH to 61.09% RH, giving a range of 0.67%RH. After EMA filtering the humidity value changed from 60.48% RH to 61.03%RH giving a smaller range of 0.55%RH. This shows that the filter reduced small fluctuations in the humidity reading.

The EMA filtered value followed the general trend of the raw humidity value, but it did not react immediately to every small change. when the raw humidity increased quickly around sample 30, the filtered value also increased, but more gradually. Similarly, when the raw humidity decreased near the end of the test, the filtered value decreased more smoothly.

3.3.3 DS18B20 Growing-Medium Temperature Sensor Results

The DS18B20 sensor was tested by collecting 100 temperature samples from the growing-medium area. The sensor produced raw digital temperature count values, and these values were converted into temperature in degrees Celsius.

During the test, the raw temperature count changed from 395 to 398. After conversion, the growing-medium temperature stayed between 24.68 °C and 24.87 °C. The average raw temperature was approximately 24.78 °C. This shows that the growing-medium temperature was stable during the test period.

Table 3-4 : Summery of DS18B20 sensor readings

Parameter	Minimum	Maximum	Average
Raw temperature count	395	398	396.57
Raw temperature	24.68 °C	24.87 °C	24.78 °C
Average filtered temperature	24.66 °C	24.87 °C	24.78 °C
EMA filtered temperature	24.66 °C	24.85 °C	24.78 °C

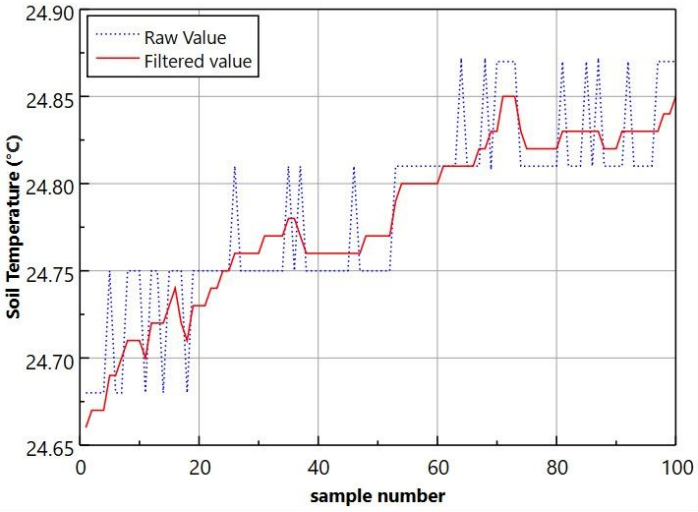


Figure 3.11 - Soil Temperature vs. sample

Filtering was applied to the DS18B20 temperature reading to reduce small changes between samples. The raw temperature changed from 24.68 °C to 24.87 °C, giving a total range of 0.19 °C. This shows that the raw DS18B20 reading was already stable.

The average filter and EMA filter made the temperature line smoother. The EMA filtered value changed from 24.66 °C to 24.85 °C. It followed the same general trend as the raw temperature but it changed more gradually. This can be seen in the graph where the raw temperature line changes in small steps, while the EMA line moves more smoothly. The filtering result shows that the DS18B20 did not have large noise during the test.

3.3.4 BH1750FVI Light Sensor Results

The BH1750FVI sensor was tested by collecting 100 light intensity samples. The sensor measured the light level reaching the plant area and gave the output in lux. During the test, the raw light intensity values changed from 10064.1 lux to 10230.0 lux. The average raw light intensity was approximately 10138.8 lux. This shows that the light level was mostly stable during the test, but small fluctuations were still present.

Table 3-5 : Summery of BH1750FVI sensor readings

Parameter	Minimum (lux)	Maximum (lux)	Average (lux)
Raw light intensity	10064.1	10230.0	10138.8
Filtered light intensity	10077.5	10213.8	10137.1
EMA light intensity	10099.8	10180.0	10135.7

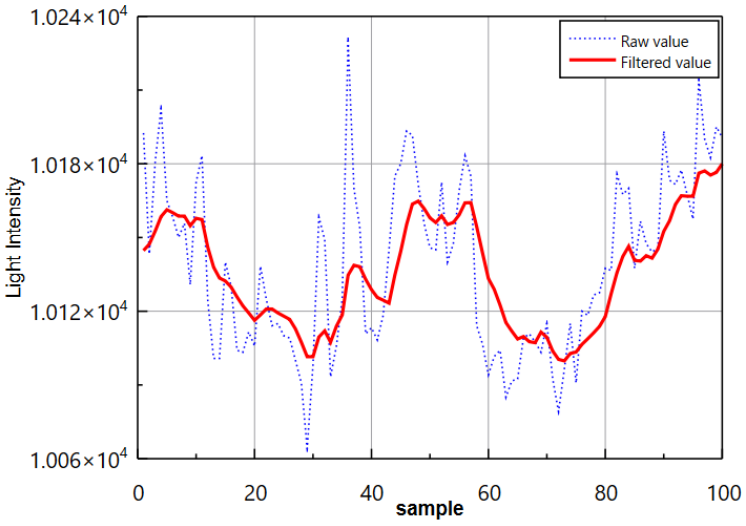


Figure 3.12 : Light intensity (lux) vs. sample

Filtering was applied to the BH1750FVI light readings because the raw lux values showed small but frequent changes. The raw light intensity changed from 10064.1 lx to 10230.0 lx, giving a total range of 165.9 lx.

After filtering, the final EMA light value changed from 10099.8 lx to 10180.0 lx. This gave a smaller range of 80.2 lx. This shows that the filter reduced the variation in the light reading by smoothing the sudden changes.

The raw light line changes quickly between samples. The filtered line follows the same general trend, but it moves more smoothly. This is useful because one sudden light reading should not strongly affect the final value sent to the grow-light unit.

The filtering result shows that the BH1750FVI sensor gave a usable light reading, but filtering improved the stability of the output.

3.4 IR uplink analysis

The purpose of this test was to check whether the control node could transmit the prepared IR feedback packet to the receiver under different distance and angle conditions. The IR feedback link was tested using different distances and angles. The result was evaluated using packet error rate.

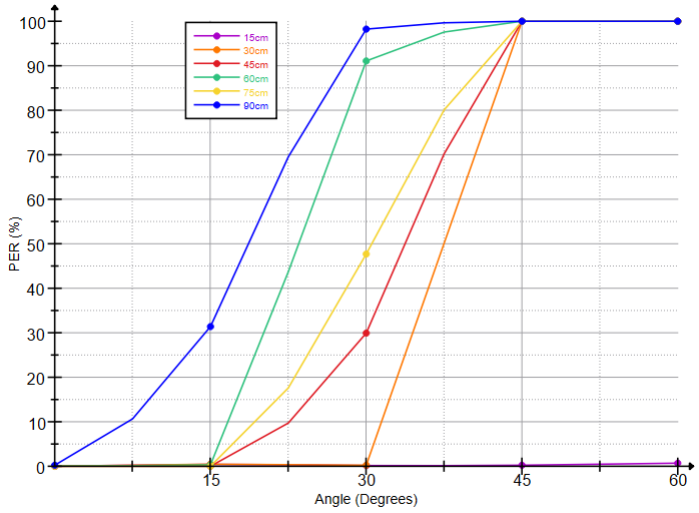


Figure 3.13 : PER (%) vs. Angle (Degree)

For each test condition, 100 packets were transmitted. This packet transmission was repeated 13 times, giving a total of 1300 transmitted packets per setup. Since each packet

contained 80 bits, each setup involved 104,000 transmitted bits. The test was repeated for different distance and angle combinations.

3.4. 1 Effect on distance

At 0° alignment, the IR transmitter was pointed directly towards the receiver. Under this condition, the IR feedback link worked reliably at all tested distances from 15 cm to 90 cm. This shows that the transmitted IR signal was strong enough when the transmitter and receiver were properly aligned. The 38 kHz modulated signal was detected by the demodulating IR receiver, and the receiver validation process accepted the packets. This confirms that the selected NEC style IR transmission method is suitable for direct line of sight communication inside the prototype system.

At 15 cm, the IR link worked well for all tested angles. This was the best distance condition because the transmitter was close to the receiver. At 30 cm, the link worked well at smaller angles, but packet failure occurred at larger angles. This shows that the signal was strong enough at short angular offsets, but not strong enough when the transmitter was turned too far away from the receiver. At 45 cm, the link worked well at 0° and 15°. However, packet errors increased at 30° and the link failed at larger angles. This shows that angular tolerance reduced when the distance increased. At 60 cm, 75 cm and 90 cm, the link became more sensitive to transmitter angle. Packet errors increased when the transmitter angle increased. Therefore, longer distance conditions required better alignment between the transmitter and receiver.

These results agree with the behavior of indoor IR links, where received signal strength depends on distance, alignment, field of view, and the optical path [1]. When the distance increases, less optical power reaches the receiver.

3.4. 2 Effect on angle

The transmitter angle had a clear effect on packet reception. At 0°, the transmitter directly faced the receiver, so this was the best operating position.

At 15°, the link still worked well at most distances, but the result became weaker at the longest distance. This shows that small angular changes are acceptable only when the

transmitter and receiver are not too far apart. At 30°, the link became unreliable at longer distances. This shows that the receiver did not receive enough optical power when the transmitter was not closely aligned. At 45° and 60°, most packets failed except at the shortest distance. This happened because the IR LED is directional. When the receiver is outside the main radiation direction, less optical power reaches the receiver. The IR LED used in the transmitter has a limited viewing angle, so the transmitted optical intensity reduces when the receiver is outside the main radiation direction [2].

The results prove that the proposed IR feedback link is reliable when it is used as a short-range line-of-sight communication method. The result also shows that repeated transmission can improve the chance of successful reception, but it cannot fully solve poor physical alignment.

CHAPTER 4. Conclusions and Future work

4.1 Conclusions

This research presented the design and development of a VLC-based passive-powered sensing and control node for indoor farming applications. The proposed system was designed to reduce wiring complexity, reduce battery dependency, and support local sensing and actuator-control functions using an optical communication and energy-harvesting approach. The developed prototype used a solar panel as a dual-purpose optical interface for both energy harvesting and VLC signal reception, following the Simultaneous Lightwave Information and Power Transfer concept.

The proposed control-node architecture successfully integrated the main hardware and firmware blocks required for local indoor farming control. The solar-panel output was separated into DC and AC paths using a bias-tee interface. The DC component was directed to the energy-harvesting path, while the AC component was filtered, amplified, converted into a digital pulse waveform, and decoded by the MSP430G2553 microcontroller.

Binary Frequency Shift Keying was selected as the final VLC modulation method and final VLC downlink used 3 kHz for binary 0, 5 kHz for binary 1, and 10 kHz for the idle state. The VLC system was tested at 50 bps, 100 bps, and 200 bps. The 200 bps configuration was selected for the final implementation because it reduced the command-frame transmission time while maintaining acceptable packet error performance. The results showed that the 200 bps VLC link achieved reliable command-frame reception over the tested operating range, with low packet error rate values up to 150 cm. The use of CRC validation ensured that corrupted frames were rejected instead of being accepted as valid actuator-control commands.

The energy harvesting system was evaluated using solar panel V-I and P-V characterization and supercapacitor charging tests. The results showed that the available solar panel power depends strongly on the received illumination level. Under the higher illumination condition, the solar panel produced a higher short-circuit current and maximum output power than under the lower illumination condition. The supercapacitor charging results also confirmed that the charging time increases as the distance from the light source increases and as the LED duty cycle decreases. During practical operation, the node worked

as a cyclic energy harvesting system, where the supercapacitor charged until the regulated output was enabled, powered the control electronics during active operation, and then recharged again after the voltage dropped below the usable level.

The sensor subsystem was developed to measure five indoor farming parameters using four sensors. The SHT3X sensor measured air temperature and relative humidity, the capacitive soil moisture sensor measured growing-medium moisture, the DS18B20 sensor measured growing-medium temperature, and the BH1750FVI sensor measured light intensity. Calibration and filtering methods were applied to improve the stability of the readings before they were used for feedback transmission and actuator control decisions. The results showed that the selected sensors were suitable for integration with the MSP430-based control node and could provide useful environmental information for strawberry-based controlled-environment cultivation.

The actuator control subsystem was also implemented and tested at the control signal level. Since the actuator loads require higher current than the passive-powered node can provide, the harvested energy was used only for the low power electronics, while the actuator loads were powered separately through the driver circuit. This separation allowed the control node to remain passive-powered while still supporting practical actuator control functions. The developed system provided control outputs for ventilation, humidification, irrigation, and thermoelectric cooling-related operation.

An infrared uplink was implemented to provide acknowledgement and feedback from the control node to the grow-light unit. The IR uplink used NEC-style timing with a 38 kHz carrier and a custom feedback frame containing sensor and actuator-status information. The IR tests showed that the feedback link operated reliably under direct line-of-sight alignment. The results also showed that the IR link is sensitive to transmitter angle and distance, which is expected for a directional infrared LED and demodulating IR receiver. Therefore, proper physical alignment is important for reliable IR feedback communication.

Overall, the developed prototype demonstrated that a passive-powered control node can combine solar-panel-based VLC reception, optical energy harvesting, low-power microcontroller operation, sensor measurement, actuator-control signal generation, and IR feedback communication in a single embedded system. The results confirm the feasibility

of using a VLC-based passive-powered control node for low-data-rate indoor farming sensing and control applications.

4.2 Limitations

1. The solar panel used as the VLC receiver. The solar panel is suitable for energy harvesting and low-data-rate signal reception, but its bandwidth is limited compared with a dedicated photodiode. Therefore, the VLC downlink was limited to low-kHz Binary FSK frequencies and low-data-rate command transmission. This is acceptable for control commands.
2. The harvested energy was sufficient for the low power electronics under suitable illumination conditions, but it was not sufficient to power high current actuator loads directly. Therefore, the actuator loads were powered separately, and the passive powered node provided only logic level control signals.
3. Under some illumination and distance conditions, the node cannot operate continuously because the harvested power is lower than the active-mode power consumption. In such cases, the node must wait until the supercapacitor recharges to the required voltage before the next active operation.
4. The IR uplink worked well under direct line-of-sight conditions, but the link performance reduced when the transmitter angle increased or when the distance increased. Therefore, receiver placement and transmitter alignment must be considered in practical deployment.
5. The BH1750FVI sensor measures illuminance in lux, which is useful for monitoring relative light level variation. However, it does not directly measure PPFD, which is more directly related to plant photosynthetic light response. Therefore, the light sensing part of the system should be improved in future work if accurate plant growth lighting evaluation is required.

4.3 Future Work

Future work can improve the proposed system in several directions.

The VLC receiver front-end can be improved to increase sensitivity, data rate and operating range by optimizing the solar panel size, selecting a panel with better frequency response,

improving the amplifier and comparator stages, and reducing the loading effect of the energy harvesting path on the communication signal.

The VLC communication protocol can be improved by adding stronger error detection, or forward-error-correction methods.

The energy harvesting and power management section can be optimized further. Lower power sensors, a lower power analog front-end, improved sleep-mode firmware, and a more efficient harvesting configuration could reduce the recharge time and increase the operating frequency of the control node. Different supercapacitor sizes can also be evaluated to balance charging time, stored energy, and physical size.

The IR uplink can be improved by using a better optical alignment method, multiple IR LEDs, a wider-angle IR transmitter, or multiple receiver positions. These improvements would increase the tolerance to node placement and make the feedback link more reliable in a practical indoor farming setup.

The sensor subsystem can be improved by adding a PPFD sensor for more accurate plant-light measurement. Additional sensors such as carbon dioxide, electrical conductivity, pH, or water-level sensors can also be integrated depending on the crop and growing method.

The actuator control section can be extended from simple ON/OFF control to closed-loop control. For example, fan speed control, proportional irrigation timing, humidifier duty-cycle control, and more controlled Peltier cooling operation can be implemented to improve environmental stability.

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